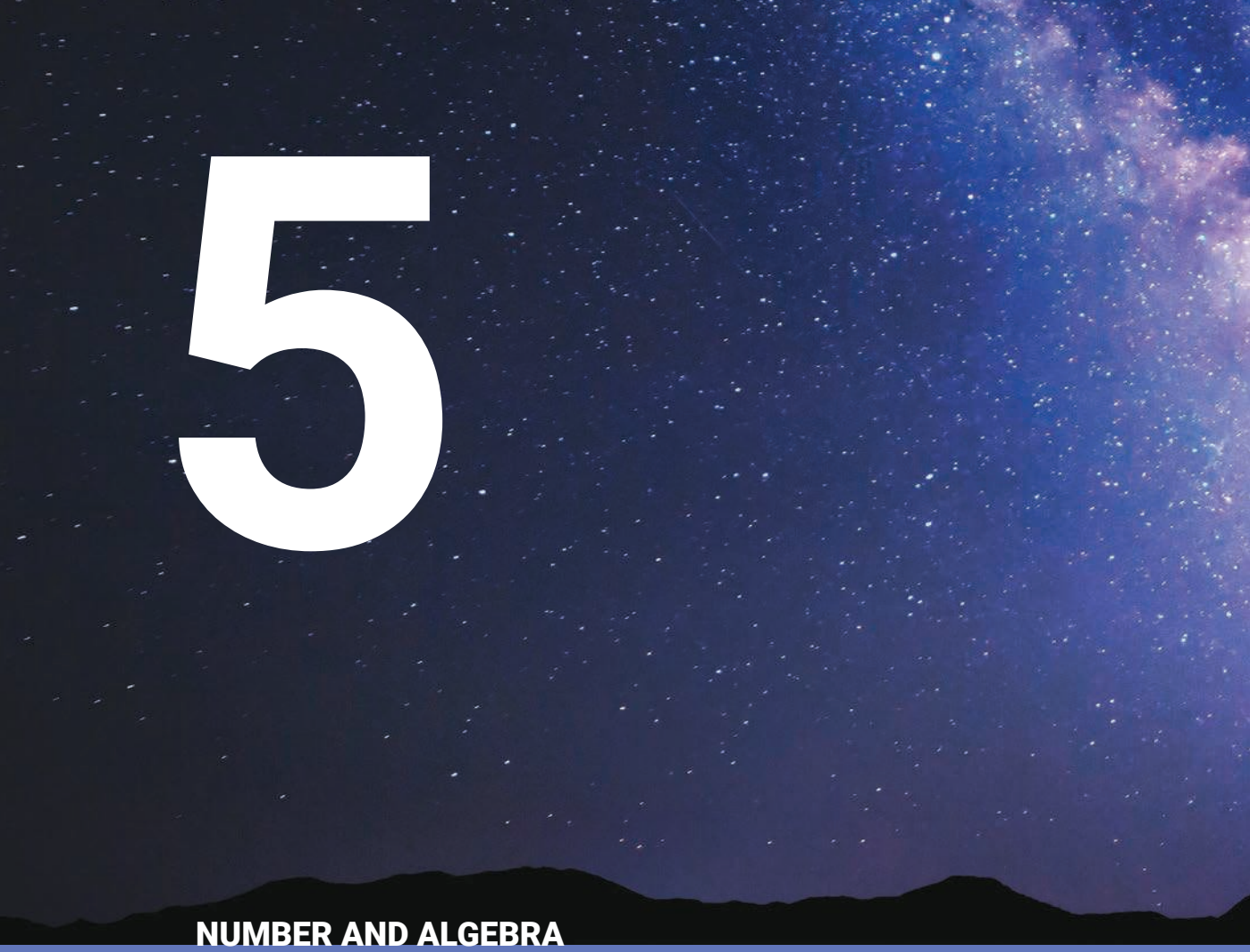


5

NUMBER AND ALGEBRA

INDICES

The speed of light is about 300 000 000 metres per second. In one year, light travels approximately 9 460 000 000 000 km and this distance is called a light year.
Light from the stars travels for many years before it is seen on Earth. The Milky Way contains over 100 billion stars and the approximate diameter of our solar system is 10 000 000 000 km. On a smaller scale, the mass of a human cell is about 0.000 000 000 001 kg.
Powers or indices provide a way to work easily with very large and very small numbers.



Chapter outline

	Working mathematically				
5.01 Multiplying and dividing terms with the same base	U	F		R	
5.02 Power of a power	U	F		R	
5.03 Powers of products and quotients	U	F		R	
5.04 The zero index	U	F		R	
5.05 Negative indices	U	F		R	C
5.06 Fractional indices $\frac{1}{n}$*	U	F		R	C
5.07 Fractional indices $\frac{m}{n}$*	U	F		R	C
5.08 Summary of the index laws	U	F		R	C
5.09 Significant figures	U	F		R	C
5.10 Scientific notation	U	F		R	C
5.11 Scientific notation on a calculator	U	F	PS	R	C

*PATH

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Wordbank

base A number that is raised to a power, meaning it is multiplied by itself repeatedly, for example, in 2^5, the base is 2.

index laws Rules for simplifying algebraic expressions involving powers of the same base, for example, $a^m \div a^n = a^{m-n}$.

index notation A way of writing repeated multiplication using indices (powers), in the form a^n, for example, $2 \times 2 \times 2 \times 2 \times 2$ in index notation is 2^5.

negative power A power that is a negative number, as in the term 3^{-2}.

power (or index or exponent) The number of times a base appears in a repeated multiplication, for example, in 2^5, the power is 5.

scientific notation A shorter way of writing very large or very small numbers using powers of 10. For example, 9 460 000 000 000 in scientific notation is 9.46×10^{12}.

significant figures Meaningful digits in a numeral that tell 'how many'. For example, 28 000 000 has 2 significant figures: 2 and 8.

Quiz
Wordbank 5

Videos (11):

SkillCheck Powers, indices and exponents

5.01, 5.02, 5.03 Simplifying with the index laws

5.02 Power of a power

5.04 Index laws 1 • Index laws 2 • The zero index

5.05 Negative indices

5.06 Fractional indices 1

5.07 Fractional indices 2

5.09 Significant figures

5.10 Scientific notation

Twig videos (3):

5.09 Rounding: Snails vs rockets

5.10 What does the Internet weigh?

5.11 Binary: the computer language

Quizzes (5):

- Wordbank 5
- SkillCheck 5
- Mental skills 5
- Language of maths 5
- Test yourself 5

Skillsheet (1):

SkillCheck Indices

Worksheets (9):

SkillCheck Powers review

5.05 Power calculations

5.08 Index laws review

5.09 Significant figures

5.10 Scientific notation 1 • Scientific notation 2

5.11 Scientific notation problems • Binary number system

Mind map Indices (Advanced)

Puzzles (5):

5.02 Indices puzzle

5.08 Indices squaresaw (Advanced)

5.10 Scientific notation puzzle

5.11 Scientific notation: accomplishing great things

Language of maths Indices crossword

Technology (1):

5.10 Scientific notation

Spreadsheet (1):

5.10 Scientific notation

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- apply index laws to numerical and algebraic expressions with integer indices
- interpret and use zero and negative integer indices
- (PATH) interpret and use fractional indices
- round numbers to a significant figures
- interpret, write and order numbers in scientific notation
- interpret and use scientific notation on a calculator
- solve problems involving scientific notation

SkillCheck ANSWERS ON P. 621

1 For each term:

- **i** state the **base**
- **ii** state the index
- **iii** write the expression in words.

a 3^5 **b** 4^8 **c** h^5 **d** 7^y

2 Evaluate each product.

a 5×5 **b** $7 \times 7 \times 7$ **c** $10 \times 10 \times 10 \times 10$

d $2 \times 2 \times 2 \times 2 \times 2 \times 2$ **e** $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$ **f** 9^2

g 3^4 **h** 6^3 **i** 4^4

j $\left(\frac{1}{2}\right)^3$ **k** $\left(\frac{3}{4}\right)^2$ **l** 5^3

3 Express each repeated multiplication in index notation.

a $2 \times 2 \times 2 \times 2 \times 2$ **b** $3 \times 3 \times 3 \times 3 \times 7 \times 7 \times 7$

c $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 8 \times 8$ **d** $\frac{3}{7} \times \frac{3}{7} \times \frac{3}{7} \times \frac{3}{7}$

e $10 \times x \times x \times x \times x \times x$ **f** $6 \times 6 \times 6 \times k \times k$

g $x \times y \times x \times y \times x$ **h** $a \times b \times b \times b \times a$

i $5 \times n \times 5 \times n \times n$ **j** $q \times p \times q \times p \times q \times q$

4 Write each term in expanded form.

a 9^3 **b** 7^2 **c** d^5 **d** k^2 **e** $\left(\frac{4}{5}\right)^3$

5 Evaluate each expression.

a $4^2 \times 4^3$ **b** $10^6 \div 10^2$ **c** $(3^3)^2$ **d** 6^0

e 9^1 **f** $5^5 \times 5$ **g** $2^4 \div 2$ **h** $(-8)^2$

6 For each equation, find the missing power.

a $8 = 2^{\square}$ **b** $81 = 3^{\square}$ **c** $216 = 6^{\square}$ **d** $144 = 12^{\square}$

e $4096 = 4^{\square}$ **f** $2401 = 7^{\square}$ **g** $64 = 2^{\square}$ **h** $625 = 5^{\square}$

Quiz SkillCheck 5

Worksheet Powers review

Skillsheet Indices

Video Powers, indices and exponents

5

Investigation

Multiplying and dividing terms with powers

1 Write each expression in expanded form, then evaluate it.

a **i** $2^2 \times 2^3$ **ii** 2^5

b **i** $3^4 \times 3^3$ **ii** 3^7

c **i** $4^3 \times 4^3$ **ii** 4^6

d **i** $5^5 \times 5^3$ **ii** 5^8

2 What do you notice about each pair of answers in question **1**?

3 Is it true that $2^4 \times 2^6 = 2^{10}$? Give a reason for your answer.

4 Determine whether each equation is true (T) or false (F). Justify your answer.

a $2^5 \times 2^5 = 2^{10}$ **b** $6^3 \times 6^7 = 6^{21}$

c $4^3 \times 4^9 = 4^{27}$ **d** $3^5 \times 3^{10} = 3^{15}$

5 Write in words and as a formula the rule for multiplying a^m and a^n, 2 terms with the same base.

6 Use the rule to copy and complete each equation.

a $5^4 \times 5^2 = 5 \ldots$ **b** $4^5 \times 4^3 = 4 \ldots$ **c** $10^5 \times 10^7 = \ldots$

d $9^3 \times 9^2 = \ldots$ **e** $n^3 \times n^8 = \ldots$ **f** $p^3 \times p^7 = \ldots$

7 Use a calculator to evaluate each expression.

a **i** $3^6 \div 3^3$ **ii** 3^3

b **i** $2^8 \div 2^6$ **ii** 2^2

c **i** $5^8 \div 5^3$ **ii** 5^5

d **i** $10^8 \div 10^4$ **ii** 10^4

8 What do you notice about each pair of answers in question **7**?

9 Is it true that $4^8 \div 4^6 = 4^2$? Give a reason for your answer.

10 Use a calculator to determine whether each equation is true (T) or false (F).

a $3^{10} \div 3^6 = 3^4$ **b** $4^8 \div 4^2 = 4^4$

c $2^{12} \div 2^3 = 2^4$ **d** $6^{10} \div 6^5 = 6^5$

11 Write in words and as a formula the rule for dividing a^m and a^n, 2 terms with the same base.

12 Use the rule to copy and complete each equation.

a $2^6 \div 2^3 = 2^{\cdots}$ **b** $10^8 \div 10^6 = 10^{\cdots}$ **c** $3^7 \div 3^2 =$

d $4^{11} \div 4^6 = \ldots$ **e** $x^8 \div x^5 = \ldots$ **f** $g^{12} \div g^{10} = \ldots$

Multiplying and dividing terms with the same base

5.01

Consider $5^4 \times 5^3 = (5 \times 5 \times 5 \times 5) \times (5 \times 5 \times 5)$

$= 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$

$= 5^7$

$\therefore 5^4 \times 5^3 = 5^{4+3}$

$= 5^7$



Video
Simplifying with the index laws

Multiplying terms with the same base

When multiplying terms with the same base, add the powers:

$$a^m \times a^n = a^{m+n}$$

The rule above is called an **index law**.

Index is another name for **power**. The plural of index is **indices** (pronounced 'in-de-sees').

Proof:

$a^m \times a^n = \underbrace{a \times a \times \cdots \times a}_{m \text{ factors}} \times \underbrace{a \times a \times \cdots \times a}_{n \text{ factors}}$

$= \underbrace{a \times a \times \cdots \times a}_{(m+n) \text{ factors}}$

$= a^{m+n}$

Example 1

Simplify each expression, writing the answer in **index notation**.

a $8^4 \times 8^5$ **b** 10×10^3 **c** $d^3 \times d^5$

d $4m^2 \times 3m^6$ **e** $3r^2t \times 6rt^3$

Solution

a $8^4 \times 8^5 = 8^{4+5}$

$= 8^9$

b $10 \times 10^3 = 10^1 \times 10^3$

$= 10^{1+3}$

$= 10^4$

c $d^3 \times d^5 = d^{3+5}$

$= d^8$

d $4m^2 \times 3m^6 = (4 \times 3) \times (m^2 \times m^6)$

$= 12m^{2+6}$

$= 12m^8$

e $3r^2t \times 6rt^3 = 3r^2t^1 \times 6r^1t^3$

$= (3 \times 6) \times (r^2 \times r^1) \times (t^1 \times t^3)$

$= 18r^{2+1}t^{1+3}$

$= 18r^3t^4$

Consider $5^6 \div 5^4 = \dfrac{5^6}{5^4}$

$$= \frac{\cancel{5}\times\cancel{5}\times\cancel{5}\times\cancel{5}\times 5\times 5}{\cancel{5}\times\cancel{5}\times\cancel{5}\times\cancel{5}}$$

$$= 5\times 5$$

$$= 5^2$$

$\therefore 5^6 \div 5^4 = 5^{6-4}$

$= 5^2$

Dividing terms with the same base

When dividing terms with the same base, subtract the powers:

$$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$$

This is another **index law**.

Proof:

$a^m \div a^n = \dfrac{a^m}{a^n}$

$= \dfrac{\cancel{a}\times\cancel{a}\times a\times a\times a\times\cdots\times a}{\cancel{a}\times\cancel{a}\times a\times\cdots\times a}$ $\left(\dfrac{m \text{ factors}}{n \text{ factors}}\right)$

$= a\times a\times \ldots \times a$ $(m-n)$ factors

$= a^{m-n}$

Example 2

Simplify each expression, writing the answer in index notation.

a $8^5 \div 8^3$ **b** $\dfrac{10^8}{10}$ **c** $d^{20} \div d^4$

d $20w^{10} \div 5w^2$ **e** $\dfrac{8x^3y^7}{24x^2y}$

Solution

a $8^5 \div 8^3 = 8^{5-3}$

$= 8^2$

b $\dfrac{10^8}{10} = \dfrac{10^8}{10^1}$

$= 10^{8-1}$

$= 10^7$

c $d^{20} \div d^4 = d^{20-4}$

$= d^{16}$

d $20w^{10} \div 5w^2 = \dfrac{\cancel{20}^{4}w^{10}}{\cancel{5}^{1}w^2}$

$= 4w^{10-2}$

$= 4w^8$

e $\dfrac{8x^3y^7}{24x^2y^1} = \dfrac{{}^{1}\cancel{8}x^{3-2}y^{7-1}}{{}_{3}\cancel{24}}$

$= \dfrac{xy^6}{3}$

Example 3

Simplify each expression, writing the answer in index notation.

a $5^3 \times 5^7 \times 5^4$ **b** $2^7 \times 2^3 \div 2^5$ **c** $m^7 \times m^6 \div m^4$

d $3e^5 \times 2e^4 \times 5e^2$ **e** $a^{12} \div a^4 \div a^5$ **f** $12y^4 \times 4y^3 \div 20y^5$

Solution

a $5^3 \times 5^7 \times 5^4 = 5^{3+7+4}$
$= 5^{14}$

b $2^7 \times 2^3 \div 2^5 = 2^{10} \div 2^5$
$= 2^5$

c $m^7 \times m^6 \div m^4 = m^{13} \div m^4$
$= m^9$

d $3e^5 \times 2e^4 \times 5e^2 = (3 \times 2 \times 5) \times (e^5 \times e^4 \times e^2)$
$= 30e^{11}$

e $a^{12} \div a^4 \div a^5 = a^{12-4-5}$
$= a^3$

f $12y^4 \times 4y^3 \div 20y^5 = \dfrac{12y^4 \times 4y^3}{20y^5}$
$= \dfrac{48y^{7-5}}{20}$
$= \dfrac{12y^2}{5}$

EXERCISE 5.01 ANSWERS ON P. 621

Multiplying and dividing terms with the same base U F R

1 Which expression is equal to $5^{12} \times 5^3$? Select the correct answer **A**, **B**, **C** or **D**.

A 5^9 **B** 5^{15} **C** 25^{15} **D** 25^{36}

EXAMPLE 1

2 Simplify each expression, writing the answer in index notation. R

a $10^3 \times 10^2$ **b** 2×2^4 **c** $3^2 \times 3^5$

d $7^4 \times 7$ **e** $8 \times 8^3 \times 8^4$ **f** $5^4 \times 5 \times 5^4$

g $6 \times 6^2 \times 6^3 \times 6^4$ **h** $4^4 \times 4^4 \times 4^4$ **i** $3^4 \times 3^0 \times 3^7$

j $x \times x^4$ **k** $g^4 \times g^4$ **l** $w^7 \times w$

m $b^3 \times b^{10}$ **n** $p^{10} \times p^{10}$ **o** $r \times r$

p $y^5 \times y^7$ **q** $m^{15} \times m^4$ **r** $n^8 \times n^2$

3 Which expression is equal to $10^4 \times 10$? Select **A**, **B**, **C** or **D**.

A 100^5 **B** 100^4 **C** 10^4 **D** 10^5

4 Which expression is equal to $5^{12} \div 5^3$? Select **A**, **B**, **C** or **D**.

A 5^4 **B** 5^9 **C** 1^4 **D** 1^9

EXAMPLE 2

5 Simplify each expression, writing the answer in index notation. R

a $10^7 \div 10^5$ **b** $8^5 \div 8$ **c** $20^{15} \div 20^5$ **d** $\dfrac{5^8}{5^2}$

e $\dfrac{9^{12}}{9^3}$ **f** $\dfrac{2^{27}}{2^3}$ **g** $7^4 \div 7^3$ **h** $\dfrac{2^{20}}{2}$

Foundation Standard Complex

i $11^4 \div 11^4$ **j** $p^{15} \div p^{10}$ **k** $n^7 \div n$ **l** $w^{24} \div w^6$

m $\frac{h^{20}}{h^4}$ **n** $\frac{y^8}{y^2}$ **o** $\frac{a^{12}}{a^4}$ **p** $b^{16} \div b^{15}$

q $\frac{w^{25}}{w}$ **r** $m^{16} \div m^{16}$

6 Which expression is equal to $10^4 \div 10$? Select **A**, **B**, **C** or **D**.

A 10^4 **B** 1^4 **C** 1^3 **D** 10^3

EXAMPLE 3

7 Which expression is equal to $6^7 \times 6^2 \times 6^4$?

A 216^{13} **B** 6^{56} **C** 6^{13} **D** 216^{56}

8 Which expression is equal to $2^7 \div 2^2 \times 2^4$?

A 2^9 **B** 2^{13} **C** 8^{13} **D** 4^9

9 Simplify each expression. R

a $3p^2 \times 2p^5$ **b** $4y^{10} \times 3y^2$ **c** $6m \times 3m^8$

d $h^3 \times 5h^8$ **e** $3q \times 8q^8$ **f** $2a^2 \times 5a^5$

g $5n^8t \times 6n^8t^4$ **h** $2ab^3 \times 15ab$ **i** $3e^4g^3 \times e^6g^2$

j $8p^4m^5 \times 4p^3m^5$ **k** $16qr^8 \times 3q^7$ **l** $9u^3v \times 6uv^2w^8$

10 Simplify each expression. R

a $10y^{15} \div 5y^3$ **b** $20w^9 \div 4w^3$ **c** $24r^8 \div 3r$

d $\frac{30x^4}{x^3}$ **e** $\frac{10m^{10}}{2m}$ **f** $\frac{4g^{12}}{8g^6}$

g $14d^4h^{10} \div 7hd^2$ **h** $15x^6y^8 \div 15xy^4$ **i** $6e^{25}d^{40} \div 18e^5d^4$

j $\frac{12q^5t^4}{16q^4t^3}$ **k** $\frac{45a^{10}b^8}{5a^5}$ **l** $\frac{36pq^3r^5}{24qr}$

11 Which expression is equal to $5d^7 \times 6d^2 \div 15d^4$? Select **A**, **B**, **C** or **D**.

A $2d^{13}$ **B** $2d^5$ **C** $15d^5$ **D** $5d^5$

12 Simplify each expression, writing the answer in index notation. R

a $3^3 \times 3^5 \times 3^2$ **b** $10^{15} \div 10^7 \div 10^2$ **c** $7^6 \times 7^8 \div 7^{10}$

d $8^9 \div 8^8 \times 8^4$ **e** $a^5 \times a^5 \times a^4$ **f** $m^3 \times m^8 \div m^{10}$

g $w^9 \div w^6 \times w^{10}$ **h** $x \times x^7 \div x^4$ **i** $p^{12} \div p \div p^6$

j $2c^5 \times 5c^8 \times 3c$ **k** $6y^4 \times 4y^8 \div 8y^4$ **l** $5d^2 \times 12d^4 \times 2d^5$

m $8q^7 \times 3q^3 \div 12q^8$ **n** $24g^{11} \div 8g^8 \div g$ **o** $6h^{10} \div h^8 \times 5h^{10}$

p $3v^5 \times 6v^4 \div 12v^6$ **q** $8b^6 \times 2b \div 10b^5$ **r** $24k^5 \div 6k^2 \div 2k^2$

13 Which expression is equal to $4p^3y^5 \times 5p^7y^3 \div 8p^6y^6$? Select **A**, **B**, **C** or **D**.

A $160p^4y^2$ **B** $\frac{20p^4y}{8}$ **C** $\frac{5p^4y}{2}$ **D** $\frac{5p^4y^2}{2}$

14 Simplify each expression, writing the answer in index notation.

a $a^{-3} \times a^3$ **b** $y \times y^{-2}$ **c** $n^{\frac{1}{2}} \times n^{\frac{1}{2}}$

d $h^{-1} \times h^{-4}$ **e** $b^4 \div b^4$ **f** $\frac{r^3}{r^5}$

Foundation Standard Complex

9780170479585

Investigation

Powers of powers

1 Write each expression in expanded form, then evaluate it.

a i $(2^3)^2$ ii 2^6

b i $(3^4)^3$ ii 3^{12}

c i $(5^2)^3$ ii 5^6

d i $(2^5)^4$ ii 2^{20}

2 What do you notice about each pair of answers in question **1**?

3 Is it true that: $(2^7)^3 = 2^{21}$? Give a reason for your answer.

4 Determine whether each equation is true (T) or false (F). Justify your answer.

a $(3^5)^3 = 3^{15}$ **b** $(2^3)^2 = 2^5$ **c** $(2^{10})^4 = 2^{14}$

d $(4^2)^5 = 4^{10}$ **e** $(3^3)^6 = 3^{18}$ **f** $(5^2)^4 = 5^6$

5 Write in words and as a formula the rule for raising a^m to a power of n, that is, $(a^m)^n$.

6 Use the rule to copy and complete each equation.

a $(3^7)^2 = 3^{\cdots}$ **b** $(5^2)^6 = 5^{\cdots}$ **c** $(4^5)^2 = 4^{\cdots}$

d $(a^3)^4 = a^{\cdots}$ **e** $(8^3)^7 = \ldots$ **f** $(k^4)^6 = \ldots$

5.01

Power of a power

5.02

Consider $(5^3)^4 = 5^3 \times 5^3 \times 5^3 \times 5^3$

$= (5 \times 5 \times 5) \times (5 \times 5 \times 5) \times (5 \times 5 \times 5) \times (5 \times 5 \times 5)$

$= 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$

$= 5^{12}$

$\therefore (5^3)^4 = 5^{3\times 4}$

$= 5^{12}$

Video
Simplifying with the index laws

Puzzle
Indices puzzle

Power of a power

When raising a term with a power to another power, multiply the powers:

$$(a^m)^n = a^{m\times n}$$

Proof:

$$(a^m)^n = \underbrace{a^m \times a^m \times \cdots \times a^m}_{n \text{ factors}}$$

$$= \underbrace{\underbrace{a\times a\times\cdots\times a}_{m \text{ factors}} \times \underbrace{a\times a\times\cdots\times a}_{m \text{ factors}} \times\cdots\times \underbrace{a\times a\times\cdots\times a}_{m \text{ factors}}}_{n \text{ lots of } m \text{ factors}}$$

$$= a^{m\times n}$$

Video
Power of a power

Example 4

Simplify each expression, writing the answer in index notation.

a $(8^5)^2$ **b** $(d^3)^5$ **c** $(2g)^4$

d $(5v^4)^3$ **e** $(-n)^6$ **f** $(-3t^4)^3$

Solution

a $(8^5)^2 = 8^{5\times 2}$
$= 8^{10}$

b $(d^3)^5 = d^{3\times 5}$
$= d^{15}$

c $(2g)^4 = 2^4 \times g^4$
$= 16g^4$

d $(5v^4)^3 = 5^3 \times (v^4)^3$
$= 125 \times v^{4\times 3}$
$= 125v^{12}$

e $(-n)^6 = (-1)^6 \times n^6$
$= 1 \times n^6$
$= n^6$

f $(-3t^4)^3 = (-3)^3 \times (t^4)^3$
$= -27 \times t^{4\times 3}$
$= -27t^{12}$

EXERCISE 5.02 ANSWERS ON P. 621

Power of a power U F R

EXAMPLE 4

1 Which expression is equal to $(10^3)^3$? Select the correct answer **A**, **B**, **C** or **D**.

A 30^3 **B** 1000^9 **C** 10^9 **D** 10^6

2 Simplify each expression, writing the answer in index notation. R

a $(4^3)^2$ **b** $(5^2)^8$ **c** $(3^3)^4$ **d** $(2^7)^4$ **e** $(2^1)^2$ **f** $(9)^3$

g $(10^0)^2$ **h** $(6^4)^5$ **i** $(5^3)^5$ **j** $(e^2)^4$ **k** $(t^5)^5$ **l** $(y^3)^7$

m $(c^1)^5$ **n** $(m^7)^5$ **o** $(y^4)^4$ **p** $(h^0)^6$ **q** $(q^6)^3$ **r** $(w^4)^1$

s $(2x)^{10}$ **t** $(5n^3)^8$ **u** $(4d^3)^3$ **v** $(-k^5)^9$ **w** $(-d^3)^4$ **x** $(2a^8)^8$

3 Which expression is equal to $(-3)^5$? Select **A**, **B**, **C** or **D**.

A -3^6 **B** -3^5 **C** 3^5 **D** -15

4 Simplify each expression. R

a $(2d^3)^4$ **b** $(5m^3)^2$ **c** $(4y^5)^2$ **d** $(3x^2)^4$

e $(5u^6)^5$ **f** $(2w^5)^3$ **g** $(10d^5)^4$ **h** $(3e)^3$

i $(2b^4)^1$ **j** $(6d^6)^2$ **k** $(3f^4)^5$ **l** $(2c^3)^{10}$

m $(-2r)^4$ **n** $(-5t)^3$ **o** $(-3m^3)^2$ **p** $(-y^3)^{12}$

q $(-x)^3$ **r** $(-m^3)^{10}$ **s** $(-4w^5)^4$ **t** $(-3f)^5$

u $(-3p^2)^3$ **v** $(-3h^5)^4$ **w** $(-10k)^2$ **x** $(-8y^3)^1$

5 Which expression is equal to $(-4m^3)^3$? Select **A**, **B**, **C** or **D**.

A $-64m^9$ **B** $-64m^6$ **C** $64m^6$ **D** $64m^9$

6 Simplify each expression. R

a $\left(d^{-1}\right)^3$ **b** $\left(x^2\right)^{-2}$ **c** $\left(w^{\frac{1}{2}}\right)^{\frac{1}{2}}$ **d** $\left(z^{-2}\right)^{-1}$ **e** $\left(u^{\frac{1}{2}}\right)^2$ **f** $\left(p^{-1}\right)^{-1}$

Foundation Standard Complex

9780170479585

Powers of products and quotients

5.03

Consider $(2 \times 5)^3 = (2 \times 5) \times (2 \times 5) \times (2 \times 5)$
$= 2 \times 2 \times 2 \times 5 \times 5 \times 5$
$= 2^3 \times 5^3$
$\therefore (2 \times 5)^3 = 2^3 \times 5^3$



Video
Simplifying with the index laws

5.03

Powers of products

When raising a product of terms to a power, raise each term to that power:

$$(ab)^n = a^n b^n$$

Proof:

$(ab)^n = \underbrace{ab \times ab \times \cdots \times ab}_{n \text{ factors}}$
$= \underbrace{a \times a \times \cdots \times a}_{n \text{ factors}} \times \underbrace{b \times b \times \cdots \times b}_{n \text{ factors}}$
$= a^n b^n$

Example 5

Simplify each expression.

a $(-2gh^2)^5$ **b** $(p^3q^4)^2$ **c** $(a^5x^2y)^3$

Solution

a $(-2gh^2)^5 = (-2)^5 \times g^5 \times (h^2)^5$
$= -32 \times g^5 \times h^{2 \times 5}$
$= -32g^5h^{10}$

b $(p^3q^4)^2 = (p^3)^2 \times (q^4)^2$
$= p^{3 \times 2} \times q^{4 \times 2}$
$= p^6q^8$

c $(a^5x^2y)^3 = (a^5)^3 \times (x^2)^3 \times y^3$
$= a^{5 \times 3} \times x^{2 \times 3} \times y^3$
$= a^{15}x^6y^3$

Consider $\left(\frac{5}{8}\right)^6 = \frac{5}{8} \times \frac{5}{8} \times \frac{5}{8} \times \frac{5}{8} \times \frac{5}{8} \times \frac{5}{8}$
$= \frac{5 \times 5 \times 5 \times 5 \times 5 \times 5}{8 \times 8 \times 8 \times 8 \times 8 \times 8}$
$= \frac{5^6}{8^6}$
$\therefore \left(\frac{5}{8}\right)^6 = \frac{5^6}{8^6}$

Powers of quotients

When raising a quotient of terms to a power, raise each term to that power:

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Proof:

$$\left(\frac{a}{b}\right)^n = \underbrace{\frac{a}{b}\times\frac{a}{b}\times\cdots\times\frac{a}{b}}_{n \text{ factors}}$$

$$= \frac{a\times a\times\cdots\times a\ (n \text{ factors})}{b\times b\times\cdots\times b\ (n \text{ factors})}$$

$$= \frac{a^n}{b^n}$$

Example 6

Simplify each expression.

a $\left(\frac{7c}{d}\right)^2$ **b** $\left(\frac{4k^2}{5}\right)^3$ **c** $\left(\frac{6m^5p^3}{m^2p}\right)^4$ **d** $\frac{2x^4y^8}{(xy^2)^3}$

Solution

a $\left(\frac{7c}{d}\right)^2 = \frac{(7c)^2}{d^2}$

$= \frac{7^2c^2}{d^2}$

$= \frac{49c^2}{d^2}$

b $\left(\frac{4k^2}{5}\right)^3 = \frac{(4k^2)^3}{5^3}$

$= \frac{4^3(k^2)^3}{125}$

$= \frac{64k^6}{125}$

c $\left(\frac{6m^5p^3}{m^2p}\right)^4 = \frac{(6m^5p^3)^4}{(m^2p)^4}$

$= \frac{6^4m^{5\times4}p^{3\times4}}{m^{2\times4}p^4}$

$= \frac{1296m^{20}p^{12}}{m^8p^4}$

$= 1296m^{20-8}p^{12-4}$

$= 1296m^{12}p^8$

OR $\left(\frac{6m^5p^3}{m^2p}\right)^4 = (6m^{5-2}p^{3-1})^4$

$= (6m^3p^2)^4$

$= 6^4m^{3\times4}p^{2\times4}$

$= 1296m^{12}p^8$

d $\frac{2x^4y^8}{(xy^2)^3} = \frac{2x^4y^8}{x^3(y^2)^3}$

$= \frac{2x^4y^8}{x^3y^6}$

$= 2x^{4-3}y^{8-6}$

$= 2xy^2$

EXERCISE 5.03 ANSWERS ON P. 621

Powers of products and quotients U F R

EXAMPLE 5

1 Which expression is equal to $(-4\times5)^2$? Select the correct answer **A**, **B**, **C** or **D**.

A 16×25 **B** -16×25 **C** -8×10 **D** 8×10

2 Simplify each expression. R

a $(ab)^3$ **b** $(x^2y)^5$ **c** $(l^3m^5)^6$ **d** $(6dp^2)^4$

e $(-8k^4y^5)^2$ **f** $(3m^2n)^5$ **g** $(ek^3)^3$ **h** $(-w^3x^4)^7$

i $(-8d^3y^5)^2$ **j** $(4b^2c^3)^4$ **k** $(-3a^3d)^3$ **l** $(2p^2q^3)^4$

3 Express each of the following in simplest form. R

a $(x^2yw^3)^2$ **b** $(h^2y^3m^5)^4$ **c** $(3d^7k^2p^3)^2$

d $(5a^3q^3r^4)^3$ **e** $(2d^4ew^3)^5$ **f** $(km^3p^2x^6)^4$

4 Which expression is equal to $\left(-\frac{3}{4}\right)^3$? Select **A**, **B**, **C** or **D**.

A $-\frac{9}{12}$ **B** $\frac{9}{12}$ **C** $\frac{27}{64}$ **D** $-\frac{27}{64}$

5 Simplify each expression. R

a $\left(\frac{6}{7}\right)^2$ **b** $\left(\frac{m}{2}\right)^3$ **c** $\left(\frac{5}{x}\right)^4$ **d** $\left(\frac{2n^5}{p}\right)^8$

e $\left(\frac{w^2}{t^3}\right)^5$ **f** $\left(\frac{m^2}{4n}\right)^4$ **g** $\left(-\frac{2}{3}\right)^4$ **h** $\left(-\frac{5h}{6}\right)^3$

i $\left(\frac{7k^4}{10}\right)^2$ **j** $\left(\frac{3r}{t^2}\right)^2$ **k** $\left(\frac{a^2b}{d^5}\right)^4$ **l** $\left(-\frac{2}{3c^2}\right)^5$

6 Simplify each expression. R

a $(2x^{10}y^{15})^3 \times 5x^2y^3$ **b** $(2x^{10}y^{15} \times 5x^2y^3)^3$ **c** $18q^5r^8 \div (3qr^2)^2$

d $(18q^5r^8 \div 3qr^2)^2$ **e** $\left(\frac{3a^5x^6}{ax}\right)^3$ **f** $\frac{3a^5x^6}{(ax)^4}$

g $(4p^3h^{10})^2 \times 2p^2h^9$ **h** $(4p^3h^{10})^2 \div 2p^2h^9$ **i** $(4p^3h^{10} \times 2p^2h^9)^2$

j $\left(\frac{2a^4d^5}{a^3d^2}\right)^3$ **k** $\left(\frac{7e^4y^2}{e^2y}\right)^2$ **l** $\frac{5x^7y^{10}}{\left(x^2y^3\right)^3}$

m $\frac{6h^9k^{12}}{\left(2h^2k^3\right)^3}$ **n** $(3m^5w^3)^4 \div 6m^9w^5$ **o** $\left(\frac{3c^5d^2}{c^2d}\right)^4$

EXAMPLE 6

5.03

Investigation

The power of 0

What is the value of a number raised to a power of 0, for example, 2^0?

1 Copy and complete each table of decreasing powers. Notice the pattern in your answers.

a

Power of 2	Number
2^5	32
2^4	16
2^3	
2^2	
2^1	
2^0	

b

Power of 3	Number
3^5	243
3^4	
3^3	
3^2	
3^1	
3^0	

Foundation Standard Complex

2 Simplify each expression in index notation.

a	$3^4 \times 3^0$	**b**	$5^2 \times 5^0$	**c**	$2^0 \times 2^7$
d	$7^0 \times 7^3$	**e**	$4^5 \times 4^0$	**f**	$5^0 \times 5^7$
g	$2^5 \div 2^0$	**h**	$3^5 \div 3^0$	**i**	$4^2 \div 4^0$
j	$9^3 \div 9^0$	**k**	$5^6 \div 5^0$	**l**	$8^4 \div 8^0$

3 Any number will remain unchanged when multiplied by what?

4 Any number will remain unchanged when divided by what?

5 What is the answer when any number is raised to the power of 0, that is, a^0? Justify your answer.

5.04 The zero index

Videos
Index laws 1
Index laws 2

Consider $5^3 \div 5^3 = \frac{5^3}{5^3}$
$= 1$ Any number divided by itself equals 1.

But also $5^3 \div 5^3 = 5^{3-3}$
$= 5^0$

$\therefore 5^0 = 1$.

Power of 0

Any number raised to the power of 0 is equal to 1.

$$a^0 = 1$$

Proof:

$a^m \div a^m = 1$ Any number divided by itself equals 1.

But also $a^m \div a^m = a^{m-m}$
$= a^0$

$\therefore\ a^0 = 1$.

Video
The zero index

Example 7

Simplify each expression.

a	11^0	**b**	$(-8)^0$	**c**	g^0
d	$(3r)^0$	**e**	$3r^0$	**f**	-8^0

Solution

a $11^0 = 1$

b $(-8)^0 = 1$

c $g^0 = 1$

d $(3r)^0 = 1$

e $3r^0 = 3 \times r^0$
$= 3 \times 1$
$= 3$

f $-8^0 = -1 \times 8^0$
$= -1 \times 1$
$= -1$

EXERCISE 5.04 ANSWERS ON P. 622

The zero index U F R

EXAMPLE 7

1 Simplify each expression. R

a	2^0	**b**	$(-2)^0$	**c**	-2^0	**d**	$(-m)^0$
e	$-m^0$	**f**	$(4a)^0$	**g**	$\left(\frac{2}{3}\right)^0$	**h**	$7x^0$
i	-1000^0	**j**	$(p+3)^0$	**k**	$\left(\frac{p}{3}\right)^0$	**l**	$-2b^0$
m	$(9k)^0$	**n**	$(x^2y)^0$	**o**	$(xyw)^0$	**p**	$(-ab)^0$
q	$(6r)^0$	**r**	$-(6r)^0$	**s**	$-6r^0$	**t**	$6(-r)^0$
u	$(cd)^0$	**v**	$-(7x^2)^0$	**w**	$-3(a^2b^3)^0$	**x**	$(-5v^5w^4)^0$

2 Simplify each expression. R

a	$7^0 + 2^0$	**b**	$12^0 - 5^0$	**c**	$2m^0 + (2m)^0$
d	$5w^0 - (5w)^0$	**e**	$(3a)^0 + 6a^0$	**f**	$(8a)^0 - 6x^0$
g	$(5y)^0 - 4$	**h**	$(5y)^0 - 4^0$	**i**	$3^0 \times 5^0$
j	$3^2 \times 5^0$	**k**	$\left(\frac{3}{4}\right)^0 + \frac{3}{4}y^0$	**l**	$\left(\frac{1}{3}\right)^0 + \left(\frac{1}{3}d\right)^0$
m	$2w^0 \times 3p^0$	**n**	$12u^0 \div 3$	**o**	$(5d^{\,0})^3$
p	$8b^0 - (3b^0)^2$	**q**	$\frac{12p^0}{(2p)^0}$	**r**	$6n^3 \div 2n^3$
s	$\frac{12q^5}{36q^5}$	**t**	$(3x^3)^3 \div x^9$	**u**	$(5h^4x^3)^0 - 4(h^3x)^0$
v	$4m^0 + 7$	**w**	$9c^0 + 2d^0$	**x**	$3 - 2(8h)^0$

5.04

Mental skills 5: Maths without calculators ANSWERS ON P. 622

Quiz
Mental skills 5

Adding or multiplying in any order

Numbers can be added or multiplied in any order. We can use this property to make our calculations simpler.

1 Study each example.

a $19 + 5 + 5 + 1 = (19 + 1) + (5 + 5)$
$= 20 + 10$
$= 30$

b $13 + 8 + 20 + 27 + 80 = (13 + 27) + (20 + 80) + 8$
$= 40 + 100 + 8$
$= 148$

c $2 \times 36 \times 5 = (2 \times 5) \times 36$
$= 10 \times 36$
$= 360$

d $25 \times 11 \times 4 \times 7 = (25 \times 4) \times (11 \times 7)$
$= 100 \times 77$
$= 7700$

Foundation Standard Complex

2 Now evaluate each sum.

a $45 + 16 + 45 + 4 + 7$ **b** $38 + 600 + 50 + 12 + 40$

c $18 + 91 + 9 + 20$ **d** $75 + 33 + 7 + 25$

e $24 + 16 + 80 + 44 + 10$ **f** $56 + 5 + 20 + 15 + 4$

g $100 + 36 + 200 + 10 + 90$ **h** $54 + 27 + 9 + 16 + 3$

i $70 + 50 + 30 + 25 + 25$ **j** $32 + 120 + 40 + 80 + 40$

3 Now evaluate each product.

a $8 \times 4 \times 5$ **b** $50 \times 7 \times 2$ **c** $3 \times 5 \times 6$

d $5 \times 11 \times 40$ **e** $12 \times 2 \times 3$ **f** $2 \times 4 \times 25 \times 8$

g $3 \times 20 \times 7 \times 5$ **h** $6 \times 8 \times 5 \times 2$ **i** $2 \times 3 \times 2 \times 11$

Investigation

Negative powers

What is the value of a number raised to a negative power, for example, 2^{-1} or 2^{-2}?

1 Copy and complete each table showing decreasing powers. Notice the pattern in your answers.

a

Power of 2	Number
2^3	8
2^2	4
2^1	
2^0	
2^{-1}	
2^{-2}	
2^{-3}	

b

Power of 10	Number
10^3	1000
10^2	
10^1	
10^0	
10^{-1}	
10^{-2}	
10^{-3}	

2 Copy and complete this table showing decreasing powers in expanded form. Notice the pattern in your answers.

a

Power of 5	Expanded form
3^3	$3 \times 3 \times 3$
3^2	3×3
3^1	3
3^0	1
3^{-1}	$\frac{1}{3}$
3^{-2}	$\frac{1}{3 \times 3} = \frac{1}{3^2}$
3^{-3}	$\frac{1}{3 \times 3 \times 3} = \frac{1}{3^3}$
3^{-4}	
3^{-5}	

b

Power of 5	Expanded form
5^3	$5 \times 5 \times 5$
5^2	
5^1	
5^0	
5^{-1}	
5^{-2}	
5^{-3}	
5^{-4}	
5^{-5}	

3 If $3^{-2} = \frac{1}{3^2}$ and $5^{-3} = \frac{1}{5^3}$, then write each negative power in a similar way.

a 4^{-1} **b** 7^{-4} **c** 2^{-6}

4 Simplify each expression in index notation.

a $10^4 \div 10^7$ **b** $2^3 \div 2^8$ **c** $3^4 \div 3^5$

d $5^2 \div 5^8$ **e** $a^4 \div a^6$ **f** $a \div a^4$

5 Consider $\frac{10^4}{10^7} = \frac{\cancel{10} \times \cancel{10} \times \cancel{10} \times \cancel{10}}{\cancel{10} \times \cancel{10} \times \cancel{10} \times \cancel{10} \times 10 \times 10 \times 10}$

$= \frac{1}{10 \times 10 \times 10}$

$= \frac{1}{10^3}$

But also $\frac{10^4}{10^7} = 10^{4-7} = 10^{-3}$

$\therefore 10^{-3} = \frac{1}{10^3}$

Use the method above to show that:

a $\frac{2^3}{2^8} = 2^{-5} = \frac{1}{2^5}$ **b** $\frac{3^4}{3^5} = 3^{-1} = \frac{1}{3}$ **c** $\frac{5^2}{5^8} = 5^{-2} = \frac{1}{5^6}$ **d** $\frac{a^4}{a^6} = a^{-2} = \frac{1}{a^2}$

6 Write in words and as a formula the rule for raising a to a negative power $-n$, that is, a^{-n}.

Technology

Negative powers

In this activity we will discover the pattern for negative powers. We will consider base values from 2 to 10 shown in column A and indices (powers) –1, –2 and –3 shown in row 1. On a spreadsheet, the symbol for power is ^ (called a **carat,** press **SHIFT 6**) . For example, 3^{-1} is entered as **3^–1.**

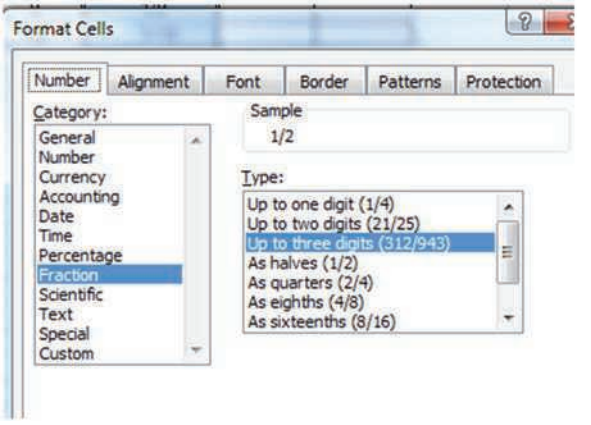
In this activity we will discover the pattern for negative powers. We will consider base values from 2 to 10 shown in column A and indices (powers) –1, –2 and –3 shown in row 1. On a spreadsheet, the symbol for power is ^ (called a **carat,** press **SHIFT 6**) . For example, 3^{-1} is entered as **3^–1.**

1 Create a spreadsheet as shown.

	A	B	C	D
1	Index	-1	-2	-3
2				
3	Base			
4	2			
5	3			
6	4			
7	5			
8	6			
9	7			
10	8			
11	9			
12	10			
13				

2 We will first examine the power of –1. In cell B4, enter **=A4^B1** to calculate 2^{-1}. B1 is an **absolute cell reference**, which ensures that the cell does not change when a formula is copied. This means that in column B, the power will always refer to cell B1 (–1) only. **Fill Down** from cell B4 to B12.

3 Use **Format cells** to set column B decimals to **Fraction** and **Up to three digits**.



4 Compare your answers in column B with the original values in column A. Can you describe the pattern when a base is raised to a power of –1?

5 Now consider powers of –2. Adapt steps from 1 to 3 for **column C**. Use **Fill Down** from cell C4 to C12.

	A	B	C	D
1	**Index**	-1	-2	-3
2				
3	**Base**			
4	2	1/2	=A4^C1	
5	3	1/3		
6	4	1/4		
7	5	1/5		

6 Compare your answers in column C with the original values in column A. Can you describe the pattern when a base is raised to a power of –2?

7 Now consider powers of –3. Adapt steps for **column D**. In cell D4, enter the formula **=A4^D1**.

Note: D12's fraction is missing as it has 4 digits in the denominator, which the spreadsheet doesn't allow for. Can you figure out what the fraction should be?

8 Compare your answers in column D with the original values in column A. Can you describe the pattern when a base is raised to a power of –3?

9 Write a rule for negative powers, given the answers you have found in this activity. Discuss with other students in your class.

Negative indices

5.05

Consider

$2^0 \div 2^3 = \frac{2^0}{2^3}$

$= \frac{1}{2^3}$

But also

$2^0 \div 2^3 = 2^{0-3}$

$= 2^{-3}$

$\therefore 2^{-3} = \frac{1}{2^3}$

Worksheet
Power calculations

5.05

Negative powers

A number raised to a negative power gives a fraction (with a numerator of 1):

$$a^{-n} = \frac{1}{a^n}$$

Proof:

$a^0 \div a^n = \frac{a^0}{a^n}$

$= \frac{1}{a^n}$

But also

$a^0 \div a^n = a^{0-n}$

$= a^{-n}$

$\therefore a^{-n} = \frac{1}{a^n}$

Example 8

Simplify each expression using a positive index (power).

a 5^{-3} **b** $3n^{-2}$ **c** $(3n)^{-2}$ **d** $p^{-2}q^3$

Solution

a $5^{-3} = \frac{1}{5^3}$

b $3n^{-2} = 3 \times n^{-2}$

$= \frac{3}{1} \times \frac{1}{n^2}$

$= \frac{3}{n^2}$

c $(3n)^{-2} = \frac{1}{(3n)^2}$

$= \frac{1}{9n^2}$

d $p^{-2}q^3 = \frac{1}{p^2} \times q^3$

$= \frac{q^3}{p^2}$

Video
Negative indices

Example 9

Write each expression with a negative index.

a $\frac{1}{c^3}$ **b** $\frac{5}{3k^2}$

Solution

a $\frac{1}{c^3} = c^{-3}$

b $\frac{5}{3k^2} = \frac{5}{3} \times \frac{1}{k^2}$

$= \frac{5}{3}k^{-2}$

PATH

Power of −1

Consider $9^{-1} = \frac{1}{9}$

$\frac{1}{9}$ is the **reciprocal** of 9.

Consider $\left(\frac{2}{3}\right)^{-1} = \frac{1}{\left(\frac{2}{3}\right)}$

$= 1 \div \frac{2}{3}$

$= 1 \times \frac{3}{2}$

$= \frac{3}{2}$

$= 1\frac{1}{2}$

Power of −1

A number **raised to a power of −1** gives its **reciprocal**.

$$a^{-1} = \frac{1}{a}$$

$$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$$

Example 10

Simplify each expression.

a $\left(\frac{4}{3}\right)^{-1}$ **b** $\left(\frac{y^2}{5}\right)^{-}$

Solution

a $\left(\frac{4}{3}\right)^{-1} = \frac{3}{4}$ **b** $\left(\frac{y^2}{5}\right)^{-1} = \frac{5}{y^2}$

Negative powers of quotients

Consider $\left(\frac{4}{5}\right)^{-2} = \frac{1}{\left(\frac{4}{5}\right)^2}$

$= \frac{1}{\frac{16}{25}}$

$= 1 \div \frac{16}{25}$

$= 1 \times \frac{25}{16}$

$= \frac{25}{16}$

$= \frac{5^2}{4^2}$

$= \left(\frac{5}{4}\right)^2$

9780170479585

Negative powers of quotients

A number **raised to a power of $-n$** gives its reciprocal raised to the power of n.

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^{n} = \frac{b^n}{a^n}$$

Proof:

$$\left(\frac{a}{b}\right)^{-n} = \frac{1}{\left(\frac{a}{b}\right)^{n}}$$
$$= \frac{1}{\frac{a^n}{b^n}}$$
$$= \frac{b^n}{a^n}$$
$$= \left(\frac{b}{a}\right)^{n}$$

Example 11

Simplify each expression.

a $\left(\frac{4}{3}\right)^{-3}$ **b** $\left(2\frac{1}{2}\right)^{-2}$ **c** $\left(\frac{3a}{b^4}\right)^{-2}$

Solution

a $\left(\frac{4}{3}\right)^{-3} = \left(\frac{3}{4}\right)^{3}$ Take the reciprocal, then cube.

$$= \frac{27}{64}$$

b $\left(2\frac{1}{2}\right)^{-2} = \left(\frac{5}{2}\right)^{-2}$

$$= \left(\frac{2}{5}\right)^{2}$$
$$= \frac{4}{25}$$

c $\left(\frac{3a}{b^4}\right)^{-2} = \left(\frac{b^4}{3a}\right)^{2}$

$$= \frac{b^8}{9a^2}$$

EXERCISE 5.05 ANSWERS ON P. 622

Negative indices UFRC

EXAMPLE 8

1 Simplify each expression using a positive index. R C

a 6^{-2}	**b** 5^{-7}	**c** 3^{-1}	**d** 10^{-2}
e g^{-5}	**f** z^{-1}	**g** n^{-3}	**h** t^{-2}
i a^{-4}	**j** 5^{-3}	**k** y^{-d}	**l** r^{-m}

2 Evaluate each expression, giving your answers in fraction form. R C

a 3^{-2}	**b** 5^{-4}	**c** 6^{-1}	**d** 7^{-2}
e 25^{-1}	**f** 2^{-7}	**g** 4^{-3}	**h** 10^{-6}
i 2^{-10}	**j** 3^{-3}	**k** 6^{-2}	**l** 9^{-4}

Foundation Standard Complex

EXAMPLE 9

3 Write each expression using a negative index. R C

a $\frac{1}{n^2}$ **b** $\frac{1}{n}$ **c** $\frac{1}{8^3}$ **d** $\frac{1}{8}$

e $\frac{1}{10^5}$ **f** $\frac{2}{a^4}$ **g** $\frac{1}{3}$ **h** $\frac{-1}{b}$

i $\frac{6}{a}$ **j** $\frac{4}{t^2}$ **k** $\frac{-2}{w^5}$ **l** $\frac{5}{d^3}$

4 Simplify each expression using positive indices. R C

a $5h^{-1}$ **b** $2b^{-5}$ **c** $3e^{-3}$ **d** $4n^{-2}$

e pb^{-2} **f** r^2s^{-4} **g** $w^{-2}y$ **h** $d^{-3}y^3$

i $(2m)^{-1}$ **j** $(xy)^{-1}$ **k** $(4h)^{-2}$ **l** $(5k)^{-3}$

m $3m^3p^{-2}$ **n** $15k^{-1}w^{-4}$ **o** $12x^{-2}y^{-3}$ **p** $12x^{-2}y^3$

q $(3h)^{-2}$ **r** $(4k)^{-3}$ **s** $(2c)^{-4}$ **t** $(8y)^{-1}$

u $4pq^{-3}$ **v** $4p^{-1}q^{-3}$ **w** vm^{-2} **x** $v^{-1}m^{-2}$

PATH

EXAMPLE 10

5 Simplify each expression. R C

a $\left(\frac{2}{7}\right)^{-1}$ **b** $\left(\frac{8}{5}\right)^{-1}$ **c** $\left(\frac{9}{10}\right)^{-1}$ **d** $\left(\frac{3}{2}\right)^{-1}$

e $\left(-\frac{3}{4}\right)^{-1}$ **f** $\left(\frac{5}{2}\right)^{-1}$ **g** $\left(\frac{x}{3}\right)^{-1}$ **h** $\left(\frac{5}{a}\right)^{-1}$

i $\left(-\frac{m}{2}\right)^{-1}$ **j** $\left(\frac{5r}{4}\right)^{-1}$ **k** $\left(\frac{2}{3z}\right)^{-1}$ **l** $\left(\frac{1}{v}\right)^{-1}$

EXAMPLE 11

6 Simplify each expression. R C

a $\left(\frac{3}{4}\right)^{-2}$ **b** $\left(\frac{2}{3}\right)^{-3}$ **c** $\left(-\frac{1}{10}\right)^{-6}$ **d** $\left(\frac{5}{2}\right)^{-3}$

e $\left(-\frac{4}{3}\right)^{-5}$ **f** $\left(\frac{5}{4}\right)^{-4}$ **g** $\left(2\frac{1}{4}\right)^{-2}$ **h** $\left(1\frac{2}{5}\right)^{-3}$

i $\left(\frac{k}{3}\right)^{-2}$ **j** $\left(-\frac{3}{x}\right)^{-1}$ **k** $\left(\frac{a^2}{5}\right)^{-4}$ **l** $\left(\frac{4}{3g^3}\right)^{-2}$

m $\left(\frac{2d}{5t}\right)^{-2}$ **n** $\left(\frac{h^2}{m^3}\right)^{-5}$ **o** $\left(-\frac{5k^7}{3p^3}\right)^{-2}$ **p** $\left(-\frac{3c^3}{4a^2}\right)^{-3}$

q $\left(\frac{5k}{2m^2n^4}\right)^{-2}$ **r** $\left(-\frac{ac^3d^2}{3}\right)^{-4}$ **s** $\left(\frac{4y^5w^2}{7k^3}\right)^{-3}$ **t** $\left(\frac{2g^3h^7}{5x^4y}\right)^{-2}$

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Fractional indices $\frac{1}{n}$

5.06

PATH

We know now the meaning of zero and negative indices, that is, a^0, a^{-1} and a^{-n}.

What is the meaning of fractional indices, that is, $a^{\frac{1}{2}}$ and $a^{\frac{1}{n}}$?

Consider $\left(25^{\frac{1}{2}}\right)^2 = 25^{\frac{1}{2}\times 2}$ Power of a power

$= 25^1$

$= 25$

but $\left(\sqrt{25}\right)^2 = 25$

$\therefore 25^{\frac{1}{2}} = \sqrt{25} = 5$

Power of $\frac{1}{2}$

Any number raised to the power of $\frac{1}{2}$ is the square root of that number:

$$a^{\frac{1}{2}} = \sqrt{a}$$

Proof:

$\left(a^{\frac{1}{2}}\right)^2 = a^{\frac{1}{2}\times 2}$

$= a^1$

$= a$

But $\left(\sqrt{a}\right)^2 = a$

$\therefore a^{\frac{1}{2}} = \sqrt{a}$

Now consider $\left(27^{\frac{1}{3}}\right)^3 = 27^{\frac{1}{3}\times 3}$ Power of a power

$= 27^1$

$= 27$

but $\left(\sqrt[3]{27}\right)^3 = 27$

$\therefore 27^{\frac{1}{3}} = \sqrt[3]{27} = 3$

Power of $\frac{1}{3}$

Any number raised to the power of $\frac{1}{3}$ is the cube root of that number:

$$a^{\frac{1}{3}} = \sqrt[3]{a}$$

PATH

Proof:

$$\left(a^{\frac{1}{3}}\right)^3 = a^{\frac{1}{3}\times 3}$$
$$= a^1$$
$$= a$$

But $\left(\sqrt[3]{a}\right)^3 = a$

$$\therefore a^{\frac{1}{3}} = \sqrt[3]{a}$$

Now consider $\left(32^{\frac{1}{5}}\right)^5 = 32^{\frac{1}{5}\times 5}$ Power of a power

$$= 32^1$$
$$= 32$$

If $\left(32^{\frac{1}{5}}\right)^5 = 32$, then $32^{\frac{1}{5}}$ is called the 5th root of 32, written $\sqrt[5]{32}$.

$$2^5 = 32$$
$$\therefore 32^{\frac{1}{5}} = \sqrt[5]{32} = 2$$

On a calculator, the nth root key is $\sqrt{\square}$ or $\sqrt[x]{\square}$, found by pressing the SHIFT or 2nd F key before pressing $x^{\square}$ or y^x respectively.

Fractional powers $\frac{1}{n}$

Any number raised to the power of $\frac{1}{n}$ is the nth root of that number:

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

Proof:

$$\left(a^{\frac{1}{n}}\right)^n = a^{\frac{1}{n}\times n}$$
$$= a^1$$
$$= a$$

but $\left(\sqrt[n]{a}\right)^n = a$

$$\therefore a^{\frac{1}{n}} = \sqrt[n]{a}$$

Example 12

Evaluate each expression.

a $900^{\frac{1}{2}}$ **b** $125^{\frac{1}{3}}$ **c** $1024^{\frac{1}{10}}$

Solution

a $900^{\frac{1}{2}} = \sqrt{900}$
$= 30$

b $125^{\frac{1}{3}} = \sqrt[3]{125}$
$= 5$

c $1024^{\frac{1}{10}} = \sqrt[10]{1024}$
$= 2$

Enter on calculator: 10 $\sqrt{\square}$ 1024 =
Note: $2^{10} = 1024$

Video
Fractional indices 1

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Example 13

Write each expression using a fractional index.

a $\sqrt{8}$ **b** $\sqrt[3]{39}$ **c** $\sqrt[4]{n}$ **d** $\sqrt[7]{ab}$ **e** $\sqrt[5]{12y}$

Solution

a $\sqrt{8} = 8^{\frac{1}{2}}$ **b** $\sqrt[3]{39} = 39^{\frac{1}{3}}$ **c** $\sqrt[4]{n} = n^{\frac{1}{4}}$

d $\sqrt[7]{ab} = (ab)^{\frac{1}{7}}$ **e** $\sqrt[5]{12y} = (12y)^{\frac{1}{5}}$

PATH

Video
Fractional indices 1

5.06

EXERCISE 5.06 ANSWERS ON P. 622

Fractional indices $\frac{1}{n}$ U F R C

1 Evaluate each expression.

EXAMPLE 12

a $25^{\frac{1}{2}}$ **b** $343^{\frac{1}{3}}$ **c** $625^{\frac{1}{2}}$ **d** $1000^{\frac{1}{3}}$

e $32^{\frac{1}{5}}$ **f** $(-0.027)^{\frac{1}{3}}$ **g** $(0.04)^{\frac{1}{2}}$ **h** $64^{\frac{1}{3}}$

i $(-8)^{\frac{1}{3}}$ **j** $(-729)^{\frac{1}{3}}$ **k** $256^{\frac{1}{8}}$ **l** $3125^{\frac{1}{5}}$

2 Write each expression using a radical (root) sign. R C

a $10^{\frac{1}{2}}$ **b** $17^{\frac{1}{3}}$ **c** $g^{\frac{1}{2}}$ **d** $m^{\frac{1}{4}}$

e $(8r)^{\frac{1}{2}}$ **f** $6h^{\frac{1}{3}}$ **g** $(3m^4)^{\frac{1}{6}}$ **h** $hk^{\frac{1}{2}}$

i $(xy^3)^{\frac{1}{7}}$ **j** $(3u^2w^5)^{\frac{1}{4}}$ **k** $d^{\frac{1}{2}}e$ **l** $15ab^{\frac{1}{3}}$

3 Write each expression using a fractional index. R C

EXAMPLE 13

a $\sqrt{5}$ **b** $\sqrt[3]{49}$ **c** $\sqrt{20}$ **d** $\sqrt[5]{400}$

e $\sqrt[6]{54}$ **f** $\sqrt[5]{80}$ **g** $\sqrt{a}$ **h** $\sqrt[3]{q}$

i $\sqrt[7]{h}$ **j** $\sqrt[6]{3w}$ **k** $\sqrt{4xy^3}$ **l** $\sqrt[5]{2m^2p^3}$

4 Evaluate each expression correct to 2 decimal places.

a $20^{\frac{1}{3}}$ **b** $215^{\frac{1}{2}}$ **c** $\sqrt[3]{36}$ **d** $\sqrt{201}$

e $(-24)^{\frac{1}{3}}$ **f** $\sqrt[4]{1111}$ **g** $\sqrt[5]{-47}$ **h** $\sqrt[6]{0.008}$

5 Simplify each expression. R

a $b^{\frac{1}{2}} \times b^{\frac{1}{2}}$ **b** $e^{\frac{1}{3}} \times e^{\frac{1}{3}} \times e^{\frac{1}{3}}$ **c** $y \times y^{\frac{1}{5}}$ **d** $m^{\frac{3}{5}} \times m^{\frac{2}{5}}$

e $2y^{\frac{1}{3}} \times 5y^{\frac{2}{3}}$ **f** $\left(-6a^{\frac{3}{2}}\right)^2$ **g** $\left(n^{12}m^4\right)^{\frac{1}{4}}$ **h** $\left(16a^2b^6\right)^{\frac{1}{2}}$

i $\left(8v^6w^9\right)^{\frac{1}{3}}$ **j** $40a^{\frac{1}{10}} \div 8a^{\frac{1}{10}}$ **k** $35x \div 5x^{\frac{1}{3}}$ **l** $36y^{\frac{1}{4}} \div 4y$

m $4m^2 \div 12m^{\frac{1}{2}}$ **n** $12k^{\frac{1}{3}} \div 15k^{\frac{1}{3}}$ **o** $\left(-2d^{\frac{1}{3}}\right)^9$ **p** $5c^{\frac{1}{3}} \times 4c^{\frac{1}{6}}$

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5.07 Fractional indices $\frac{m}{n}$

PATH

Video
Fractional indices 2

What is the meaning of fractional indices such as $a^{\frac{2}{3}}$ and $a^{\frac{3}{2}}$?

Consider $32^{\frac{3}{5}} = \left(32^{\frac{1}{5}}\right)^3$

$= \left(\sqrt[5]{32}\right)^3$

$= 2^3$

$= 8$

or consider $32^{\frac{3}{5}} = \left(32^3\right)^{\frac{1}{5}}$

$= \sqrt[5]{32^3}$

$= \sqrt[5]{32\,768}$

$= 8$

Fractional powers $\frac{m}{n}$

$$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m \text{ or } \sqrt[n]{a^m}$$

Proof:

$a^{\frac{m}{n}} = \left(a^{\frac{1}{n}}\right)^m$ or $\left(a^m\right)^{\frac{1}{n}}$

$= \left(\sqrt[n]{a}\right)^m$ or $\sqrt[n]{a^m}$

Note: Taking the root first often makes the calculation simpler

Example 14

Evaluate each expression.

a $8^{\frac{2}{3}}$ **b** $81^{\frac{5}{4}}$ **c** $64^{-\frac{1}{3}}$ **d** $16^{-\frac{3}{4}}$

Solution

a $8^{\frac{2}{3}} = \left(\sqrt[3]{8}\right)^2$

$= 2^2$

$= 4$

b $81^{\frac{5}{4}} = \left(\sqrt[4]{81}\right)^5$

$= 3^5$

$= 243$

c $64^{-\frac{1}{3}} = \frac{1}{64^{\frac{1}{3}}}$

$= \frac{1}{\sqrt[3]{64}}$

$= \frac{1}{4}$

d $16^{-\frac{3}{4}} = \frac{1}{16^{\frac{3}{4}}}$

$= \frac{1}{\left(\sqrt[4]{16}\right)^3}$

$= \frac{1}{2^3}$

$= \frac{1}{8}$

Example 15

Evaluate $300^{\frac{3}{5}}$, correct to 2 decimal places.

Solution

$300^{\frac{3}{5}} = 30.638\,870\,63...$

≈ 30.64

Enter on calculator: 300 $x^{\blacksquare}$ 3 $\frac{\blacksquare}{\blacksquare}$ 5 =

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Example 16

Write each expression using a fractional index.

a $\sqrt[4]{p^3}$ **b** $\sqrt{b^7}$ **c** $\dfrac{1}{\sqrt[5]{q^2}}$

Solution

a $\sqrt[4]{p^3} = (p^3)^{\frac{1}{4}}$
$= p^{\frac{3}{4}}$

b $\sqrt{b^7} = (b^7)^{\frac{1}{2}}$
$= b^{\frac{7}{2}}$

c $\dfrac{1}{\sqrt[5]{q^2}} = \dfrac{1}{(q^2)^{\frac{1}{5}}}$
$= \dfrac{1}{q^{\frac{2}{5}}}$
$= q^{-\frac{2}{5}}$

Example 17

Simplify each expression.

a $\sqrt[4]{(16r^2)^3}$ **b** $\left(\sqrt[3]{27k}\right)^2$ **c** $(32a^5)^{-\frac{3}{5}}$

Solution

a $\sqrt[4]{(16r^2)^3} = (16r^2)^{\frac{3}{4}}$
$= 16^{\frac{3}{4}} r^{2\times\frac{3}{4}}$
$= 8r^{\frac{3}{2}}$

b $\left(\sqrt[3]{27k}\right)^2 = (27k)^{\frac{2}{3}}$
$= 27^{\frac{2}{3}} k^{\frac{2}{3}}$
$= 9k^{\frac{2}{3}}$

c $(32a^5)^{-\frac{3}{5}} = \dfrac{1}{(32a^5)^{\frac{3}{5}}}$
$= \dfrac{1}{32^{\frac{3}{5}} a^{5\times\frac{3}{5}}}$
$= \dfrac{1}{8a^3}$

EXERCISE 5.07 ANSWERS ON P. 623

Fractional indices $\frac{m}{n}$ U F R C

1 Evaluate each expression. R C

EXAMPLE 14

a $4^{\frac{3}{2}}$ **b** $8^{\frac{5}{3}}$ **c** $128^{\frac{5}{7}}$ **d** $27^{\frac{2}{3}}$

e $1024^{\frac{3}{5}}$ **f** $64^{\frac{4}{3}}$ **g** $32^{\frac{3}{5}}$ **h** $81^{\frac{3}{4}}$

i $1000^{\frac{2}{3}}$ **j** $125^{\frac{2}{3}}$ **k** $8^{-\frac{1}{3}}$ **l** $81^{-\frac{1}{4}}$

m $25^{-\frac{3}{2}}$ **n** $36^{-\frac{3}{2}}$ **o** $256^{-\frac{3}{4}}$ **p** $3125^{-\frac{4}{5}}$

q $1024^{-\frac{4}{5}}$ **r** $400^{-\frac{3}{2}}$ **s** $128^{-\frac{4}{7}}$ **t** $1024^{-\frac{7}{10}}$

2 Evaluate each expression correct to 2 decimal places.

EXAMPLE 15

a $15^{\frac{3}{4}}$ **b** $8^{-\frac{7}{5}}$ **c** $50^{\frac{5}{4}}$ **d** $6^{-\frac{2}{3}}$

e $100^{-\frac{3}{4}}$ **f** $16^{\frac{3}{5}}$ **g** $12^{-\frac{3}{2}}$ **h** $179^{\frac{2}{5}}$

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PATH

EXAMPLE 16

EXAMPLE 17

3 Write each expression using a fractional index. R C

a $\sqrt[7]{g^2}$ **b** $\sqrt{e^5}$ **c** $\sqrt[6]{x^{18}}$ **d** $\frac{1}{\sqrt[4]{y^{16}}}$

e $\sqrt[3]{m^5}$ **f** $\sqrt[5]{m^3}$ **g** $\frac{1}{\sqrt[4]{n^3}}$ **h** $\frac{1}{\sqrt[3]{5h^4}}$

4 Simplify each expression. R C

a $\left(16n^4\right)^{\frac{3}{4}}$ **b** $\left(8w\right)^{\frac{2}{3}}$ **c** $\sqrt[5]{\left(32d^{10}\right)^3}$

d $\left(\sqrt{64m^8}\right)^3$ **e** $\left(\sqrt[4]{81r^4}\right)^5$ **f** $\sqrt[4]{\left(81h^{12}\right)^3}$

g $\frac{1}{\left(\sqrt[3]{8s^6}\right)^4}$ **h** $\left(\frac{1}{\sqrt[5]{1024x^{15}}}\right)^2$ **i** $\left(125b^6\right)^{-\frac{2}{3}}$

j $\left(625t^2\right)^{-\frac{3}{4}}$ **k** $\left(49p^4q^{10}\right)^{\frac{3}{2}}$ **l** $\left(1000x^3y^6\right)^{-\frac{2}{3}}$

5 Find the value of y if: R C

a $x^y = \frac{x^3}{\sqrt[5]{x^7}}$ **b** $x^y = \frac{\sqrt{x^5}}{\sqrt[3]{x^2}}$ **c** $x^y = \frac{\sqrt[3]{x^8}}{\sqrt{x^3}}$

d $\sqrt[4]{x^3} = \frac{x^y}{\sqrt[5]{x^2}}$ **e** $\sqrt[7]{x^2} = \frac{\sqrt[4]{x^5}}{x^y}$ **f** $x^y = \frac{\sqrt[5]{x^9}}{\sqrt{x^3}}$

5.08 Summary of the index laws

Puzzle
Indices squaresaw (Advanced)

Worksheet
Index laws review

PATH

The index laws

$a^m \times a^n = a^{m+n}$

$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$

$(a^m)^n = a^{m \times n}$

$(ab)^n = a^n b^n$

$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

$a^0 = 1$

$a^{-1} = \frac{1}{a}$

$a^{-n} = \frac{1}{a^n}$

$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$

$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^{-n} = \frac{b^n}{a^n}$

$a^{\frac{1}{2}} = \sqrt{a}$

$a^{\frac{1}{3}} = \sqrt[3]{a}$

$a^{\frac{1}{n}} = \sqrt[n]{a}$

$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$

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EXERCISE 5.08 ANSWERS ON P. 623

Summary of the index laws U F R C

1 Simplify each expression. R

a $a^4 \times a^3$ **b** $t^8 \times t$ **c** $n^8 \div n^2$ **d** $p^3 \div p$

e $(w^2)^4$ **f** $(g^3)^6$ **g** $2b^2 \times 3b^5$ **h** $4d^7 \times 5d^6$

i $\frac{30c^{12}}{5c^8}$ **j** $(5b^4)^4$ **k** $24m^6 \div 8m^4$ **l** $(3a)^2$

2 Evaluate each expression. R

a 4^0 **b** $(-9)^0$ **c** 7×2^0 **d** $(5 \times 3)^0$

e $(-2)^3$ **f** $(-3)^2$ **g** $(5^2)^4$ **h** $11^4 \times 11^3$

i $(9^2)^0$ **j** $3^5 \div 3^2$ **k** $8^2 \div 8^5$ **l** $10^3 \div 10^3$

m $5^2 \div 5^0$ **n** $10^{-2} \div 10^2$ **o** $\left(\frac{1}{2}\right)^0$ **p** $8^{-5} \times 8^5$

3 Evaluate each expression, giving your answers in fraction form. R C

a 5^{-2} **b** 3^{-5} **c** 20^{-1} **d** 10^{-3}

4 Evaluate each expression. R C PATH

a $16^{\frac{1}{2}}$ **b** $27^{\frac{1}{3}} + 4^0$ **c** $25^{\frac{5}{2}}$ **d** $8^{\frac{1}{3}} + 4^{\frac{1}{2}}$

e $\left(8^2\right)^{\frac{1}{3}}$ **f** $\left(9^3\right)^{\frac{1}{2}}$ **g** $81^{\frac{3}{4}}$ **h** $(-32)^{\frac{3}{5}}$

5 Simplify each expression. R

a $(3mn^3)^2$ **b** $8a^2w^2 \times 5a^3w^7$ **c** $(4a^2b^5)^4$ **d** $\frac{20p^3q^8}{5p^2q^6}$

e $\left(\frac{4}{5}\right)^3$ **f** $6c^2d^{\,0}$ **g** $\frac{48u^5v^4}{16uv}$ **h** $\left(\frac{3x}{10}\right)^2$

i $(-4n^2t)^3$ **j** $\left(-\frac{2}{3}\right)^4$ **k** $\frac{7x^2y^6}{35x^5y^3}$ **l** $\left(\frac{p^5}{9y}\right)^2$

m $(2p^3q^2)^5$ **n** $\left(\frac{1}{7n}\right)^3$ **o** $-2n^0$ **p** $\frac{(a^2b)^4 \times a^3}{b^5}$

6 Simplify each expression using a positive index. R C PATH

a 8^{-7} **b** 3^{-5} **c** y^{-1} **d** x^{-3}

e $(5b)^{-2}$ **f** $5b^{-2}$ **g** $(ab)^{-1}$ **h** ab^{-1}

i $11t^{-3}$ **j** $(11t)^{-3}$ **k** p^3q^{-5} **l** mw^{-3}

m $8u^{-3}v^{-4}$ **n** $-2r^6y^{-5}$ **o** $10e^{-1}f^{\,3}$ **p** $\frac{1}{2}k^{-4}n^7$

7 Simplify each expression. R

a $\left(\frac{7}{4}\right)^{-1}$ **b** $\left(\frac{5}{2}\right)^{-1}$ **c** $\left(\frac{2}{3}\right)^{-1}$ **d** $\left(\frac{1}{7}\right)^{-1}$

e $\left(\frac{r}{8}\right)^{-1}$ **f** $\left(\frac{1}{10p}\right)^{-1}$ **g** $\left(\frac{6y}{w}\right)^{-1}$ **h** $\left(\frac{2}{5a}\right)^{-1}$

PATH

8 Write each expression using a negative index. R C

a $\frac{1}{4^3}$ **b** $\frac{1}{2}$ **c** $\frac{1}{10^4}$ **d** $\frac{1}{9^2}$

e $\frac{1}{k}$ **f** $\frac{1}{3m}$ **g** $\frac{12}{y^4}$ **h** $\frac{-4}{x^7}$

i $\frac{5}{2p^3}$ **j** $\frac{-2}{5a^6}$ **k** $\frac{1}{8w^4}$ **l** $\frac{-1}{6h^2}$

9 Simplify each expression, expressing your answer with a positive index. R C

a $q^5 \times q^{-2}$ **b** $d^{-3} \times d^7$ **c** $m^{-6} \div m^5$ **d** $y \div y^{-1}$

e $5g^3 \times 6g^{-1}$ **f** $8a^{-2} \times 3a^3$ **g** $7x^{-2} \times 4x$ **h** $\frac{64p^{-1}}{16p^2}$

i $48q \div 3q^{-2}$ **j** $\frac{5d^3}{10d^{-1}}$ **k** $2(b^{-1})^4$ **l** $(3h)^{-2}$

10 Write each expression using a fractional index. R C

a $\sqrt{5}$ **b** $\sqrt[3]{d}$ **c** $\sqrt{3y}$ **d** $\sqrt[4]{10}$

e $\left(\sqrt[3]{p}\right)^2$ **f** $\sqrt{(xy)^5}$ **g** $\sqrt[4]{(5a)^3}$ **h** $\left(\sqrt{\frac{y}{6}}\right)^5$

11 Simplify each expression. R

a $\left(\frac{4}{5a}\right)^{-3}$ **b** $\left(8c^3\right)^{\frac{2}{3}}$ **c** $\left(\frac{10}{7m}\right)^{-2}$ **d** $\left(25w^5\right)^{\frac{5}{2}}$

e $\left(\frac{49}{d^2}\right)^{\frac{3}{2}}$ **f** $\left(\frac{a^3b^9}{c^6}\right)^{\frac{2}{3}}$ **g** $\sqrt[4]{625m^6}$ **h** $\left(\frac{64}{y^3}\right)^{-\frac{2}{3}}$

i $\sqrt[5]{32m^{10}}$ **j** $\left(\frac{1}{3g^2}\right)^{-2}$ **k** $\left(16x^8\right)^{\frac{5}{4}}$ **l** $\left(\frac{2a^3}{c^2}\right)^4$

5.09 Significant figures

Worksheet
Significant figures

Video
Rounding: Snails vs rockets

A way of rounding a number is to give the most relevant or important digits of the number. For example, a crowd of 47 321 people can be written as 47 000, which is rounded to the nearest thousand, or to 2 **significant figures**.

The first significant figure in a number is the first **non-zero** digit.

Number	First significant digit	Number of significant digits
47 321	4	5
47 000	4	2
0.000 159 2	1	4
0.000 2	2	1

- When rounding to significant figures, start counting from the first digit that is not 0.
- If it is a large number, you may need to insert 0s at the end as placeholders.
- 0s at the end of a whole number or at the beginning of a decimal are **not significant:** they are necessary placeholders.

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- 0s *between* significant figures or at the end of a decimal *are* significant. For example, the significant figures are shown in bold in this table.

Number	First significant digit	Number of significant digits
809 000	8	3
0.0**20 70**	2	4

5.09

Example 18

State the number of significant figures in each number.

a 63.70 **b** 0.003 05 **c** 7600

Solution

a The 0 after 7 is significant.

∴ 63.70 has 4 significant figures.

b The first significant figure is 3, and the 0 between 3 and 5 is significant.

∴ 0.003 05 has 3 significant figures.

c The 0s after 6 are not significant.

∴ 7600 has 2 significant figures.

Example 19

Video
Significant figures

Round each number to 3 significant figures.

a 56.357 **b** 9.249 **c** 548 307

Solution

a **56.3**57 ≈ 56.4 **b** **9.24**9 ≈ 9.25 **c** **548** 307 ≈ 548 000

The 0s in part **c** are not significant, but they are placeholders that are necessary for showing the place values of the 5, 4 and 8.

Example 20

Video
Significant figures

Write each number correct to one significant figure.

a 0.007 39 **b** 0.025 **c** 0.963

Solution

a 0.00**7** 39 ≈ 0.007 **b** 0.0**2**5 ≈ 0.03 **c** 0.**9**63 ≈ 1

The 0s at the beginning of a decimal are not significant: they are placeholders.

EXERCISE 5.09 ANSWERS ON P. 623

Significant figures UFRC

EXAMPLE 18

1 State the number of significant figures in each number. R C

a	457	**b**	0.23	**c**	15 000
d	4.0004	**e**	0.0005	**f**	5000
g	0.002 07	**h**	89 072	**i**	0.040
j	76 000 000	**k**	0.000 328	**l**	169.320

EXAMPLE 19

2 Round each number to 3 significant figures. C

a	37.609	**b**	9435	**c**	168.39
d	2.813	**e**	15.99	**f**	60 522
g	1 769 000	**h**	385 764	**i**	10.2717

EXAMPLE 20

3 Write each number correct to 2 significant figures. C

a	0.0637	**b**	0.903	**c**	0.084 55
d	0.000 158	**e**	0.007 625	**f**	0.038 71
g	0.2795	**h**	0.018 944	**i**	0.3145

4 What is 45 067 853 rounded to 3 significant figures? Select the correct answer **A**, **B**, **C** or **D**.

A 45 167 853 **B** 45 100 000 **C** 45 067 900 **D** 45 070 000

5 What is 0.005 605 0 rounded to 2 significant figures? Select **A**, **B**, **C** or **D**.

A 0.01 **B** 0.010 000 0 **C** 0.0056 **D** 0.005 600 0

6 Round each number to one significant figure.

a	9.478	**b**	57.12	**c**	0.0367
d	0.007 66	**e**	0.5067	**f**	10 675
g	1856.78	**h**	0.000 28	**i**	56 239 400

7 A company makes a profit of $35 754 125. C

a Round the profit to the nearest million and state the number of significant figures in the answer.

b Round the profit to the nearest 10 million and state the number of significant figures in the answer.

8 Australia's population in 2019 was 25 498 000. To how many significant figures has this number been written? C

9 A total of 21 558 people attended a football match. Express this number to 3 significant figures. C

10 The radius of a hydrogen atom is 0.000 000 000 104 m. To how many significant figures has this number been written? Select **A**, **B**, **C** or **D**.

A 12 **B** 9 **C** 3 **D** 2

11 Evaluate each expression, correct to the number of significant figures shown in the brackets.

a $45.6 \times 8.7 - 2.75 \times 78.32$ [2]

b $15.5 - 9.87 \div 0.24 + 8.43 \times 2.4$ [1]

c $(63.73 - 27.89) \div 5.82$ [3]

d $\dfrac{63.25 + 76.03}{55.89 - 89.24}$ [4]

e $\dfrac{9.732 + 2.765}{12.27 \times 15.8}$ [1]

f $78.91 \div (23.6 + 94.7)$ [2]

g $\dfrac{1}{0.941} + \dfrac{253}{0.0076}$ [3]

h $\sqrt{84.3} \times 0.0715$ [4]

Did you know?

Big numbers

The table below lists the names of some big numbers and their meanings.

According to the Guinness Book of Records, the largest number for which there is an accepted name is the centillion, first recorded in 1852. It is equal to 10^{303}.

What is the special name for the number:

a $\mathbf{10^{100}}$**?**

b $\mathbf{10^{(10^{100})}}$**?**

Name	Numeral
million	$10^6 = 1\,000\,000$
billion	$10^9 = 1\,000\,000\,000$
trillion	10^{12}
quadrillion	10^{15}
quintillion	10^{18}
sextillion	10^{21}
septillion	10^{24}
octillion	10^{27}
nonillion	10^{30}
decillion	10^{33}

Scientific notation 5.10

Scientific notation is a short way of writing very large or very small numbers using powers of 10. It was invented in the early 20th century when scientists needed to describe very large values, such as astronomical distances, and very small values, such as the masses of atoms.

Scientific notation

Numbers written in **scientific notation** are expressed in the form

$$m \times 10^n$$

where m is a number between 1 and 10 and n is an integer.

Worksheets
Scientific notation 1
Scientific notation 2

Puzzle
Scientific notation puzzle

Technology
Scientific notation

Spreadsheet
Scientific notation

Foundation | Standard | Complex

Example 21

Express each number in scientific notation.

a 764 000 000 000 **b** 6000 **c** 0.0008 **d** 0.000 000 472

Solution

Videos
Scientific notation
What does the Internet weigh?

a Use the significant figures in the number to write a value between 1 and 10: 7.64

Count how many places the decimal point moves to the right to make 764 000 000 000.

764 000 000 000 — or count the number of places after the first significant figure, 7

11 places

$\therefore 764\,000\,000\,000 = 7.64 \times 10^{11}$

b Use the significant figures in the number to write a value between 1 and 10: 6

Count how many places the decimal point moves to the right to make 6000.

6000 — or count the number of places after the first significant figure, 6

3 places

$\therefore 6000 = 6 \times 10^{3}$

c Use the significant figures in the number to write a value between 1 and 10: 8

Count how many places the decimal point moves to the left to make 0.0008.

0.0008 — or count the number of decimal places to the first significant figure, 8

4 places

$\therefore 0.0008 = 8 \times 10^{-4}$

Note that small numbers are written with *negative* powers of 10.

d Use the significant figures in the number to write a value between 1 and 10: 4.72

Count the number of places the decimal point moves to the left to make 0.000 000 472.

0.000 000 472 — or count the number of decimal places to the first significant figure, 4

7 places

$\therefore 0.000\,000\,472 = 4.72 \times 10^{-7}$

Example 22

Express each number in decimal form.

a 2.7×10^{4} **b** 3.56×10^{-2}

Solution

a $2.7 \times 10^{4} = 2.7000$ Move the decimal point 4 places to the right.

$= 27\,000$

b $3.56 \times 10^{-2} = 0.0356$ Move the decimal point 2 places to the left.

$= 0.0356$

Example 23

a Which number is the larger: 3.65×10^{12} or 8.1×10^{12}?

b Write these numbers in **ascending order**: 4.3×10^6, 2.8×10^7, 1.9×10^7

Solution

To compare numbers in scientific notation, first compare the powers of 10.
If the powers of 10 are the same, then compare the decimal parts.

a The powers of 10 are the same. Compare the decimal parts: $8.1 > 3.65$.
$\therefore$ The larger number is 8.1×10^{12}

b Compare the powers of 10: $10^6 < 10^7$.
Then compare the 2 numbers with 10^7: $1.9 < 2.8$.
$\therefore$ The numbers in ascending order are 4.3×10^6, 1.9×10^7, 2.8×10^7.

EXERCISE 5.10 ANSWERS ON P. 623

Scientific notation U F R C

EXAMPLE 21

1 Express each number in scientific notation. R C

a	2400	**b**	786 000	**c**	55 000 000	**d**	95
e	7.8	**f**	348 000 000	**g**	59 670	**h**	15
i	3 000 000 000	**j**	80	**k**	763	**l**	10
m	0.035	**n**	0.000 076	**o**	0.8	**p**	0.0713
q	0.000 003	**r**	0.913	**s**	0.000 007 146	**t**	0.009
u	0.000 000 1	**v**	0.000 89	**w**	0.000 000 078	**x**	0.1

2 Express the value shown in each measurement in scientific notation. C



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a The world's largest mammal is the blue whale, which can weigh up to 130 000 kg.

b The diameter of an oxygen molecule is 0.000 000 29 cm.

c The thickness of a human hair is 0.000 08 m.

d Light travels at a speed of 300 000 000 m/s.

e The nearest star to Earth, excluding the Sun, is Alpha Centauri, which is 40 000 000 000 000 km away.

f The thickness of a typical piece of paper is 0.000 12 m.

g The small intestine of an adult is approximately 610 cm long.

h The diameter of a hydrogen atom is 0.000 000 000 106 m.

i The diameter of our galaxy, the Milky Way, is 770 000 000 000 000 000 000 m.

j A microsecond means 0.000 001 s.

k The Andromeda Galaxy is the most remote body visible to the naked eye, at a distance of 2 200 000 light years away.

EXAMPLE 22

3 Express each number in decimal form. R C

a	6×10^5	**b**	7.1×10^3	**c**	3.02×10^8
d	3.14×10^0	**e**	6×10^{-5}	**f**	7.1×10^{-3}
g	3.02×10^{-8}	**h**	5.9×10^{-10}	**i**	1.1×10^{12}
j	4×10^{-4}	**k**	5×10^3	**l**	4.76×10^{-4}
m	8.03×10^{-1}	**n**	6.32×10^4	**o**	1.6×10^{-2}
p	2.2×10^{-7}	**q**	9.0×10^6	**r**	1.11×10^{-1}

EXAMPLE 23

4 For each pair of numbers, write the larger one. R

a	3×10^5 or 4×10^5	**b**	8.4×10^5 or 2.7×10^6
c	8.4×10^0 or 1.3×10^7	**d**	3.6×10^{-7} or 6.3×10^{-7}
e	9.3×10^9 or 7.6×10^9	**f**	3.5×10^{-6} or 9.3×10^2
g	3.04×10^0 or 3.04×10^{-4}	**h**	4.5×10^{-5} or 3.7×10^{-7}
i	2×10^{-15} or 2×10^{-17}	**j**	6.23×10^{-5} or 9.7×10^{-5}

5 Write each set of numbers in ascending order. R

a	3.8×10^9, 7.3×10^9, 5.5×10^9	**b**	2.2×10^{-4}, 5.8×10^{-6}, 7×10^{-4}
c	3.5×10^0, 5.3×10^2, 4.9×10^2		

6 Write each set of numbers in **descending order**. R

a	6×10^5, 2.9×10^2, 8×10^2	**b**	1.2×10^{-9}, 6.3×10^2, 8.1×10^{-4}
c	4.1×10^{-1}, 9.5×10^{-1}, 6.4×10^{-3}		

5.11 Scientific notation on a calculator

Worksheet Scientific notation problems

Puzzle Scientific notation: accomplishing great things

To enter a number in scientific notation on a calculator, use the $\times 10^x$ or EXP key.

Example 24

Evaluate each expression using scientific notation.

a $(4.25 \times 10^7) \times (8.2 \times 10^6)$ **b** $(1.08 \times 10^{-15}) \div (3 \times 10^{11})$ **c** $(4.9 \times 10^7)^2$

Solution

a Enter 4.25 $\times 10^x$ 7 $\times$ 8.2 $\times 10^x$ 6 =

$(4.25 \times 10^7) \times (8.2 \times 10^6) = 3.485 \times 10^{14}$

Note that with scientific notation on a calculator, there is no need to enter brackets ( ) around the numbers.

b Enter 1.08 $\times 10^x$ $-$ 15 $\div$ 3 $\times 10^x$ 11 =

$(1.08 \times 10^{-15}) \div (3 \times 10^{11}) = 3.6 \times 10^{-27}$

c Enter 4.9 $\times 10^x$ 7 x^2 =

$(4.9 \times 10^7)^2 = 2.401 \times 10^{15}$

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Example 25

Estimate the value of each expression in scientific notation, then evaluate it correct to 3 significant figures.

a $\dfrac{9.2 \times 10^9}{2.7 \times 10^5}$ **b** $(8.5 \times 10^4) \times (6.3 \times 10^7)$ **c** $(6.08 \times 10^5)^3$

Solution

Estimate

a
$$\frac{9.2 \times 10^9}{2.7 \times 10^5} \approx \frac{9 \times 10^9}{3 \times 10^5}$$
$$= \frac{9}{3} \times \frac{10^9}{10^5}$$
$$= 3 \times 10^4$$

b
$$(8.5 \times 10^4) \times (6.3 \times 10^7) \approx (9 \times 10^4) \times (6 \times 10^7)$$
$$= (9 \times 6) \times (10^4 \times 10^7)$$
$$= 54 \times 10^{11}$$
$$= 5.4 \times 10 \times 10^{11}$$
$$= 5.4 \times 10^{12}$$

c
$$(6.08 \times 10^5)^3 \approx (6 \times 10^5)^3$$
$$= 6^3 \times (10^5)^3$$
$$= 216 \times 10^{15}$$
$$= 2.16 \times 10^2 \times 10^{15}$$
$$= 2.16 \times 10^{17}$$

Calculated answer

a
$$\frac{9.2 \times 10^9}{2.7 \times 10^5} = 34\,074.074\,07$$
$$\approx 34\,000$$
$$= 3.40 \times 10^4$$

b
$$(8.5 \times 10^4) \times (6.3 \times 10^7) = 5.355 \times 10^{12}$$
$$\approx 5.36 \times 10^{12}$$

c
$$(6.08 \times 10^5)^3 = 2.24755... \times 10^{17}$$
$$\approx 2.25 \times 10^{17}$$

EXERCISE 5.11 ANSWERS ON P. 624

Scientific notation on a calculator U F PS R C

1 Evaluate each expression using scientific notation. EXAMPLE 24

a $(2 \times 10^3) \times (3 \times 10^5)$ **b** $(8 \times 10^7) \div (4 \times 10^2)$ **c** $(2 \times 10^5)^3$

d $\sqrt{9 \times 10^{12}}$ **e** $(4 \times 10^7) \times (6 \times 10^8)$ **f** $(1 \times 10^8) \div (2 \times 10^3)$

g $(4 \times 10^3)^5$ **h** $24.08 \div (8 \times 10^6)$ **i** $\sqrt{3.969 \times 10^{19}}$

j $(2 \times 10^5)^{-2}$ **k** $\sqrt[3]{8 \times 10^{-9}}$ **l** $\dfrac{7.62 \times 10^9}{2 \times 10^{-4}}$

2 Estimate the value of each expression in scientific notation, then evaluate correct to 3 significant figures. R EXAMPLE 25

a $(5.7 \times 10^3) \times (2.3 \times 10^5)$ **b** $(8 \times 10^5) \times (3.7 \times 10^7)$

c $(9.1 \times 10^{20}) \div (3.2 \times 10^5)$ **d** $(1.2 \times 10^8)^2$

e $(7.13 \times 10^{10}) \times (9.8 \times 10^8)$ **f** $(1.9 \times 10^{11}) \div (2.1 \times 10^7)$

g $(5.85 \times 10^4)^3$ **h** $(6 \times 10^{12}) \div (2.8 \times 10^3)$

3 The human body consists of approximately 6×10^9 cells, and each cell consists of 6.3×10^9 atoms. Roughly how many atoms are there in a human body? PS R

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4 *The Complete Sherlock Holmes* is a book that is 4.5 cm thick and contains 2000 pages. Find the thickness, in *millimetres*, of one page in scientific notation. PS R

5 Evaluate each expression in scientific notation, correct to 2 significant figures.

a $(7.4 \times 10^{30}) - (3.59 \times 10^{29})$

b $(1.076 \times 10^{17}) + (2.3 \times 10^{16})$

c $\sqrt{6.6 \times 10^{27}}$

d $(7.5 \times 10^{23}) \div (3.3 \times 10^{-13})$

e $(8.17 \times 10^{16})^3$

f $\sqrt{2.69 \times 10^{45}}$

g $(7.05 \times 10^{3}) \div (3.9 \times 10^{7})$

h $\dfrac{(5.6 \times 10^{4}) \times (3.9 \times 10^{5})}{(2.3 \times 10^{7})}$

i 1595×1959

j 5^{20}

k 801^{-1}

l 3^{-10}

m 9^{9}

n $(0.7)^{-5}$

6 The Earth is 1.50×10^{8} km from the Sun and the speed of light is 3×10^{5} km/s. How long does it take for light to travel from the Sun to Earth? Express your answer in: PS R C

a seconds

b minutes and seconds

7 The Sun burns 6 million tonnes of hydrogen a second. Calculate to 2 significant figures how many tonnes of hydrogen it burns in a year (that is, 365.25 days). PS R

8 Sound travels at approximately 330 metres per second. If Mach 1 is the speed of sound, how fast is Mach 5? Convert your answer to kilometres per second to 2 significant figures. PS R C

9 The distance that light travels in one year is called a light year. If the speed of light is approximately 3×10^{5} km per second, how far to 2 significant figures does light travel in a leap year? PS R

10 A thunderstorm is occurring 30 km from where you are standing. Use the speed of light (3×10^{5} km per second) and the speed of sound (330 metres per second) to calculate in seconds how long it takes: PS R C



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a the light from the lightning to reach you

b the sound from the thunder to reach you.

11 **a** What is the largest number that can be displayed on your calculator?

b What is the smallest number that can be displayed? PS R C

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Investigation

A lifetime of heartbeats

How many times does your heart beat in an average lifetime of 80 years?

1 Work in pairs and copy this table.

Name	Trial 1	Trial 2	Average beats per minute

2 Use 2 fingers to measure your pulse. Have your partner time you for a minute. Do this twice, record your results in the table and find the average.

3 Repeat Step **2** for your partner.

4 Calculate how many times your heart (and your partner's heart) beats in the following periods. Write your answers in scientific notation, correct to 2 significant figures.

a an hour **b** a day **c** a week

d a year (use 365.25 days) **e** an average lifetime of 80 years

Did you know?

Hairy numbers



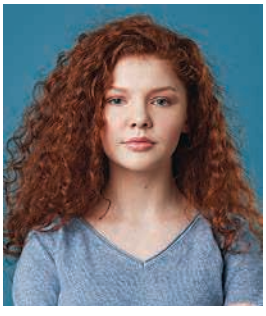
Shutterstock.com/Tracy Whiteside

Straight hair
• round follicle



Alamy Stock Photo/Stephen Flint

Wavy hair
⬬ oval follicle



Shutterstock.com/Cookie Studio

Curly hair
▬ flat follicle

There are about 110 000 hairs on your head. Each hair grows at the rate of about 1.3×10^{-3} cm per hour. A single hair lasts about 6 years. Every day you lose between 30 and 60 hairs. Each hair grows from a small depression in the skin called a follicle (a gland). After the hair falls out, the follicle rests for about 3 to 4 months before the next hair starts growing. Hair follicles are either oval, flat or round in shape. How straight, wavy or curly your hair is depends on the shape of your hair follicles.

How many hairs are on all the heads in China if its population is approximately 1.445×10^9? Answer in both scientific notation and decimal notation.

Worksheet
Binary number system

Video
Binary: the computer language

Power plus ANSWERS ON P. 624

1 Write each number in scientific notation.

a 438.2×10^{9} **b** 0.52×10^{-7} **c** 0.0004×10^{12}

d 2013×10^{-3} **e** 57.8 thousand **f** 57.8 thousandths

g 6.7 millionths **h** 3.2 billion **i** 3.2 billionths

2 Evaluate each expression.

a $\sqrt{\sqrt{81}}$ **b** $\sqrt{\sqrt{625}}$ **c** $\sqrt{\sqrt{\sqrt{256}}}$

d $\sqrt{\sqrt[5]{1024}}$ **e** $\sqrt{\sqrt{\sqrt{6561}}}$ **f** $\sqrt[4]{\sqrt{256}}$

g $\sqrt{\sqrt[3]{1000000}}$ **h** $\sqrt[3]{\sqrt{1000000}}$

3 Find values of a, m and n so that each equation is true.

a $a^{\frac{m}{n}} = 2$ **b** $a^{\frac{m}{n}} = 3$ **c** $a^{\frac{m}{n}} = 64$ **d** $a^{\frac{m}{n}} = 125$

4 Find values of a and b that solve $a^b = b^a$. What if a and b are different?

5 The terms in the pattern 3, 5, 17, 257, 65 537, ... can all be generated by a simple method, using only the numbers 1 and 2.

a What is this method?

b What is the next number in the sequence?

6 What is the last digit of each number?

a 2^{200} **b** 3^{200} **c** 7^{200}

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CHAPTER 5 REVIEW

Language of maths

ascending	base	descending	estimate
expanded form	exponent	fractional power	index
index laws	index notation	indices	negative power
product	power	quotient	reciprocal
significant figures	scientific notation	term	zero power

1 What does a power of $\frac{1}{2}$ mean?

2 Which 2 words from the list mean 'power'?

3 What is the $\times 10^x$ or EXP key on a calculator used for?

4 What is the index law for dividing terms with the same base?

5 Which digits in 0.006 701 are **significant figures**?

6 What power is associated with the **reciprocal** of a term or number?

7 What type of numbers when written in **scientific notation** have **negative** powers of 10?

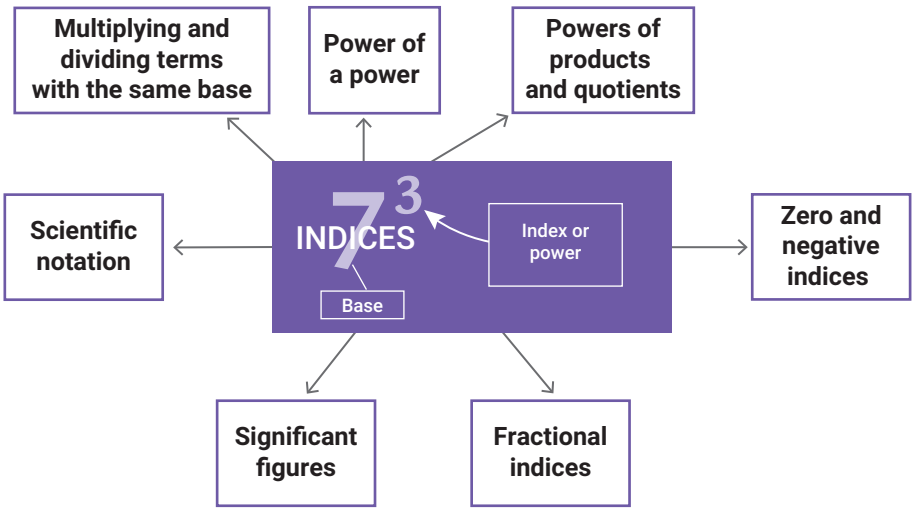
Quiz
Language of maths 5

Puzzle
Indices crossword

Topic summary

- What was this topic about? What was the main theme?
- What content was new and what was revision?
- What are the index laws?
- Write 10 questions (with solutions) that could be used in a test for this chapter.
- Include some questions that you have found difficult to answer.
- List the sections of work in this chapter that you did not understand. Follow up this work with a friend or your teacher.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Indices (Advanced)

Quiz
Test yourself 5

TEST YOURSELF 5

ANSWERS ON P. 624

5.01 **1** Simplify each expression, writing the answer in index notation.

a $10^3 \times 10^7$ **b** $4^{20} \div 4^4$ **c** $a^{12} \div a^2$

d $h^8 \times h^2$ **e** $3n^3 \times 4n$ **f** $10d^{15} \div 5d^3$

g $20m^9 \div 4m$ **h** $3v^4w^2 \times 2v^3w^5$ **i** $5x^5y^2 \times 3xy$

j $\frac{24t^8h^8}{3t^4h^2}$ **k** $\frac{p^6q^{10}}{p^2q^2}$ **l** $\frac{100a^2b^4}{5ab^2}$

5.02 **2** Simplify each expression, writing the answer in index notation.

a $(2^2)^3$ **b** $(k^5)^5$ **c** $(x^3)^4$

d $(2y^3)^{10}$ **e** $(5t^2)^2$ **f** $(10g)^3$

g $(-2)^5$ **h** $(-2k)^5$ **i** $(-5m^3)^2$

5.03 **3** Simplify each expression.

a $(ab^2)^4$ **b** $(5x^3y^2)^2$ **c** $(4t^2)^3$

d $(-4h^2g)^3$ **e** $\left(\frac{a}{7}\right)^4$ **f** $(2pqr)^5$

g $\left(\frac{3m}{2}\right)^5$ **h** $(-3np^2)^4$ **i** $\left(\frac{2a^7}{b}\right)^4$

j $(4t^4u^5)^3 \times 8t^2u$ **k** $\left(\frac{b^8y^6}{8b^2y}\right)^3$ **l** $45c^6d^8 \div (3cd^2)^2$

5.04 **4** Simplify each expression.

a 7^0 **b** $(-23)^0$ **c** e^0

d $(-k)^0$ **e** $-m^0$ **f** $8g^0h$

g $(5wy)^0$ **h** $\left(\frac{2p^5}{3}\right)^0$ **i** $\frac{2p^0}{3}$

j $15u^2 \div 20u^0$ **k** $(13a)^0 - 13a^0$ **l** $4^0 \times 5 \times (6x)^0$

5.05 **5** Simplify each expression using a positive index.

a 8^{-3} **b** 19^{-2} **c** x^{-1} **d** p^{-5}

e $(4m)^{-1}$ **f** $(3g^2)^{-3}$ **g** $\left(\frac{2}{b}\right)^{-4}$ **h** $5b^{-1}$

i $-2x^{-4}$ **j** $\left(\frac{3}{5a^2c}\right)^{-1}$ **k** $c^{-4}d^2$ **l** $\left(\frac{100}{9}\right)^{-1}$

5.05 **6** Write each expression using a negative index.

a $\frac{1}{10^3}$ **b** $\frac{1}{r^5}$ **c** $\frac{1}{kg}$ **d** $\frac{3}{b}$

PATH

5.05 **7** Simplify $\left(-\frac{8}{3x}\right)^{-2}$ using a positive index.

5.06 **8** Write each expression using a radical (root) sign.

a $q^{\frac{1}{3}}$ **b** $u^{\frac{1}{2}}$ **c** $(2q)^{\frac{1}{3}}$ **d** $(ar)^{\frac{1}{2}}$

5.07 **9** Evaluate each expression.

a $64^{\frac{2}{3}}$ **b** $(-32)^{\frac{3}{5}}$ **c** $36^{-\frac{3}{2}}$

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10 Simplify each expression.

a $\left(125d^{15}\right)^{\frac{4}{3}}$ **b** $\left(16y^{20}\right)^{\frac{1}{4}}$ **c** $\left(32x^{8}\right)^{\frac{2}{5}}$ **d** $\sqrt[3]{-64n^{12}}$

e $\left(\sqrt[5]{32k^{10}}\right)^{3}$ **f** $\left(\frac{1}{\sqrt{4g}}\right)^{5}$ **g** $\left(\sqrt[4]{81a^{2}}\right)^{6}$ **h** $\left(\frac{\left(2\sqrt[3]{8y}\right)^{4}}{10}\right)^{-2}$

11 Simplify each expression using a positive index.

a $\left(\frac{3}{2m}\right)^{-2}$ **b** $3q^{-5} \times 2q^{3}$ **c** $\left(36w^{2}\right)^{\frac{3}{2}}$ **d** $24k^{3} \div 8k^{-5}$

e $\sqrt[3]{125y^{9}}$ **f** $\left(\frac{5d}{3k}\right)^{-3}$ **g** $36p^{-4} \div 27p^{-5}$ **h** $\left(\frac{8}{m^{3}}\right)^{-\frac{2}{3}}$

i $4a^{2}b^{-5} \times 8ab^{4}$ **j** $\left(2d^{-2}\right)^{-3}$ **k** $\left(\frac{16}{81w^{8}}\right)^{-\frac{1}{4}}$ **l** $\sqrt[5]{32g^{5}h^{15}}$

m $\left(\frac{3x^{2}}{u^{5}}\right)^{4}$ **n** $\left(\frac{8n^{6}}{27c^{9}}\right)^{-\frac{2}{3}}$ **o** $8h^{3} \div 12h^{8}$ **p** $\frac{(2c^{4}e^{3})^{4}}{\sqrt[3]{c^{3}e^{6}}}$

12 Round each value correct to the number of significant figures shown in the brackets.

a 8.5678 [2] **b** 15 712 [3] **c** 476 [1]

d 0.007 126 6 [4] **e** 0.9045 [3] **f** 301 378 [2]

g 4805.28 [3] **h** 0.000 87 [1] **i** 67 000 000 [1]

13 Express each number in scientific notation.

a 37 000 **b** 0.61 **c** 250 000

d 0.000 49 **e** 13 **f** 0.000 000 000 08

14 Express each number in decimal form.

a 8.1×10^{5} **b** 6×10^{7} **c** 5.182×10^{0}

d 2.03×10^{-4} **e** 3×10^{-8} **f** 9.475×10^{-2}

15 Write these numbers in ascending order: 3×10^{3}, 9.1×10^{-8}, 2.4×10^{3}.

16 Evaluate each expression using scientific notation.

a $(3.65 \times 10^{12}) \times (7.4 \times 10^{8})$ **b** $(1.47 \times 10^{10}) \div (3.72 \times 10^{4})$

c $(8.41 \times 10^{5})^{3}$ **d** $\sqrt{6.25 \times 10^{-8}}$

17 Estimate the value of each expression in scientific notation, then evaluate correct to 2 significant figures.

a $(8.9 \times 10^{9}) \times (1.1 \times 10^{7})$ **b** $(9.3 \times 10^{15}) \div (4 \times 10^{2})$

c $(3.1 \times 10^{4})^{2}$

PATH

5.07

5.08

5.09

5.10

5.10

5.10

5.11

5.11

TEST YOURSELF 5

Foundation Standard Complex